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EQUILIBRIUM OF CONCURRENT COPLANAR FORCE SYSTEMS

Definition:-

If a system of forces acting on a body, keeps the body in a state of rest or in a state of uniform motion along a straightline, then the system of forces is said to be in equilibrium.

ALTERNATIVELY, if the resultant of the force system is zero, then, the force system is said to be in equilibrium.

Conditions for Equilibrium of Concurrent Coplanar Force System

A coplanar concurrent force system will be in equilibrium if it satisfies the following two conditions:

i) $\sum F_x = 0$; and ii) $\sum F_y = 0$



i.e. Algebraic sum of components of all the forces of the system, along two mutually perpendicular directions, is ZERO.

Graphical conditions for Equilibrium

Triangle Law: If three forces are in equilibrium, then, they form a closed triangle when represented in a Tip to Tail arrangement, as shown in Fig 1.(a).

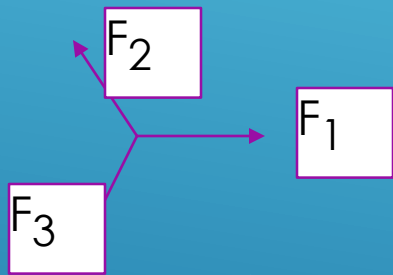
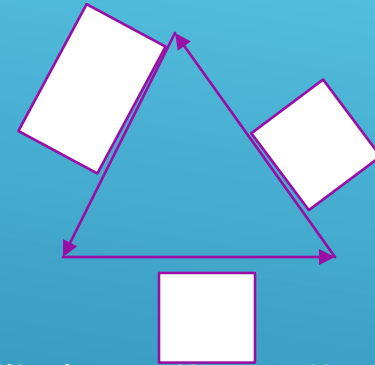


Fig 1.(a)



Polygonal Law: If more than three forces are in equilibrium, then, they form a closed polygon when represented in a Tip to Tail arrangement, as shown in Fig 1.(b).

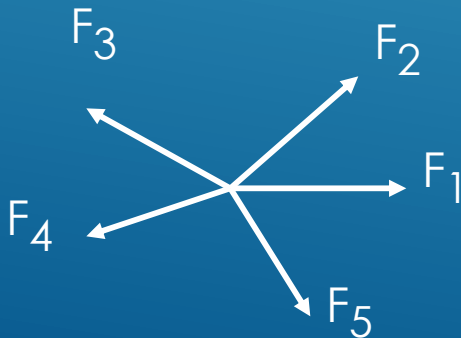
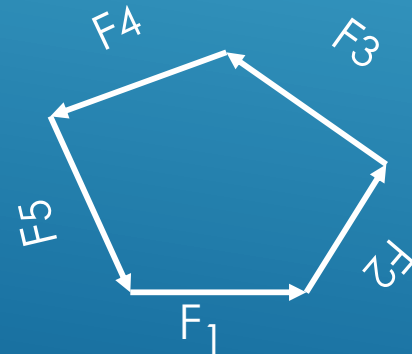


Fig 1.(b)



If a system of Three forces is in equilibrium, then, each force of the system is proportional to sine of the angle between the other two forces (and constant of proportionality is the same for all the forces). Thus, with reference to Fig(2), we have,

$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$$

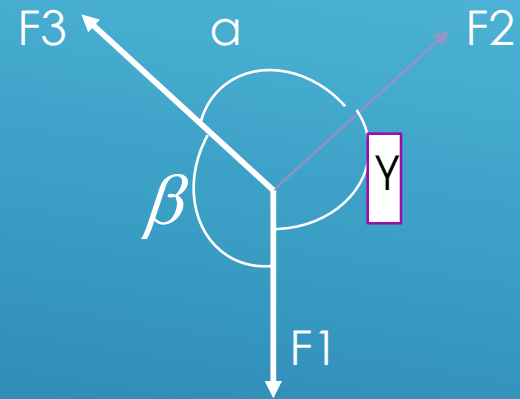


Fig (2)

Note: While using Lami's theorem, all the three forces should be either directed away or all directed towards the point of concurrence.

SPACE DIAGRAMS & FREE BODY DIAGRAMS

Space Diagram (SPD): The sketch showing the physical conditions of the problem, like, the nature of supports provided; size, shape and location of various bodies; forces applied on the bodies, etc., is known as space diagram.

Eg. Fig 3(a) is a space diagram

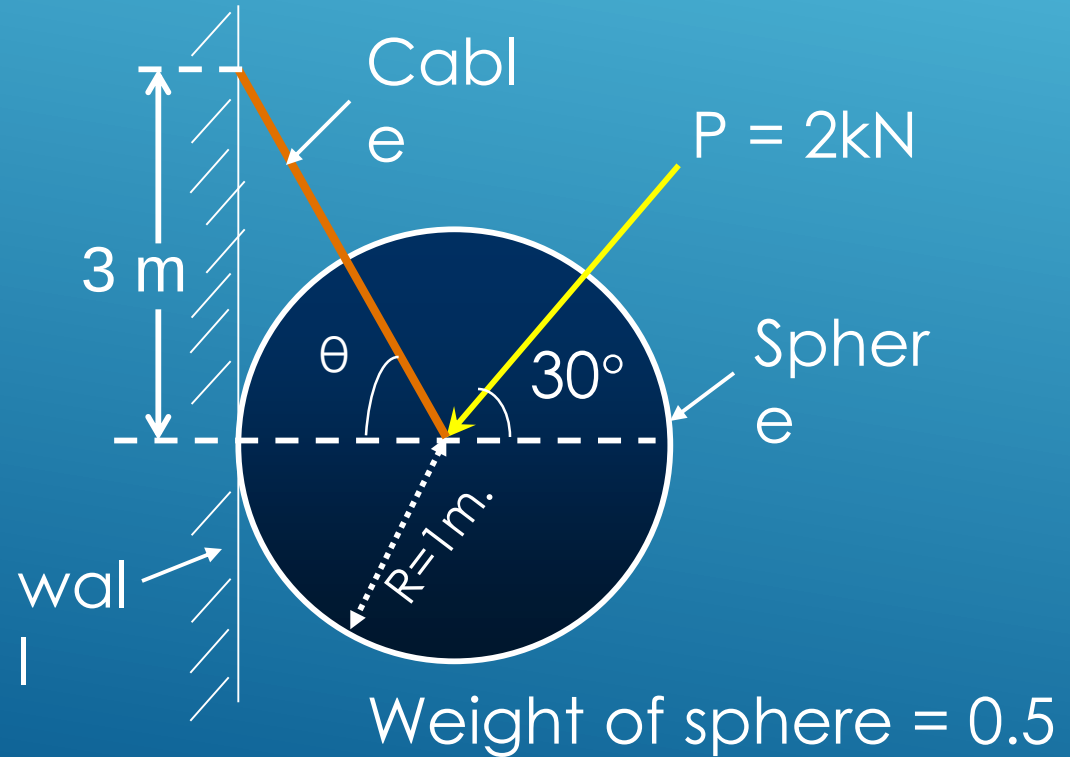
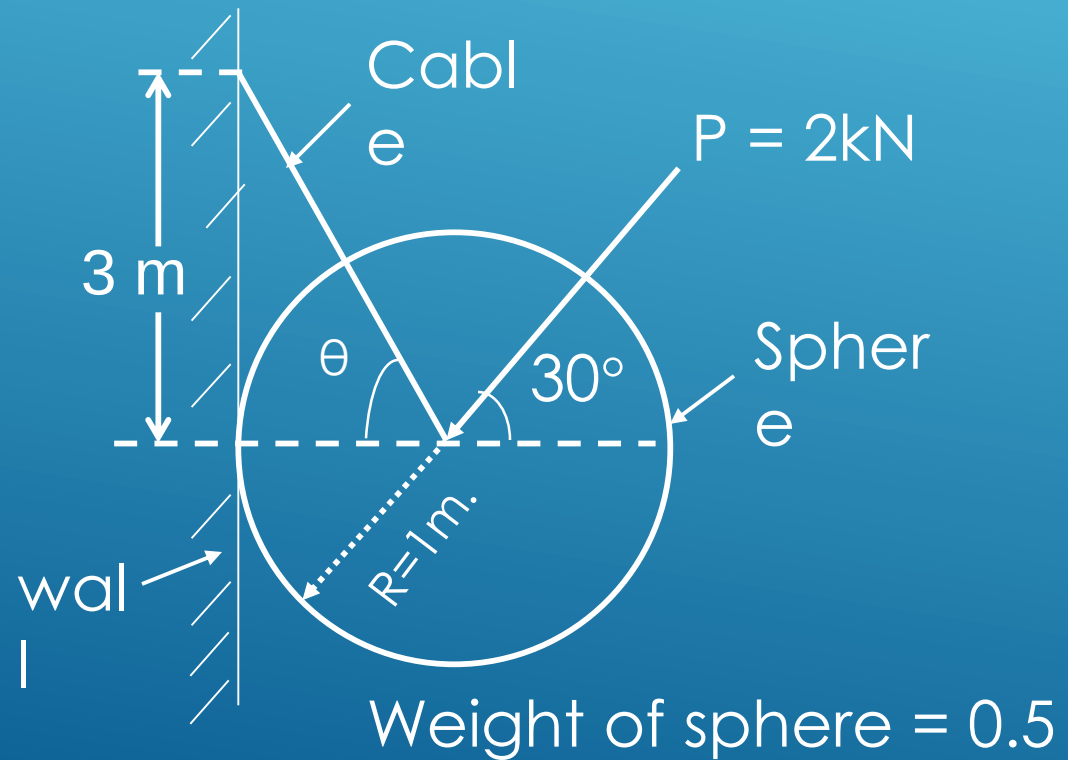


Fig 3 (a) SPD

Free Body Diagram(FBD) : It is an isolated diagram of the body being analyzed (called free body), in which, the body is shown freed from all its supports and contacting bodies/surfaces. Instead of the supports and contacting bodies/surfaces, the reactive forces exerted by them on the free body is shown, along with all other applied forces.

Eg. Fig 3(b) is a Free Body Diagram of Fig 3(a).

Fig 3 (a) SPD



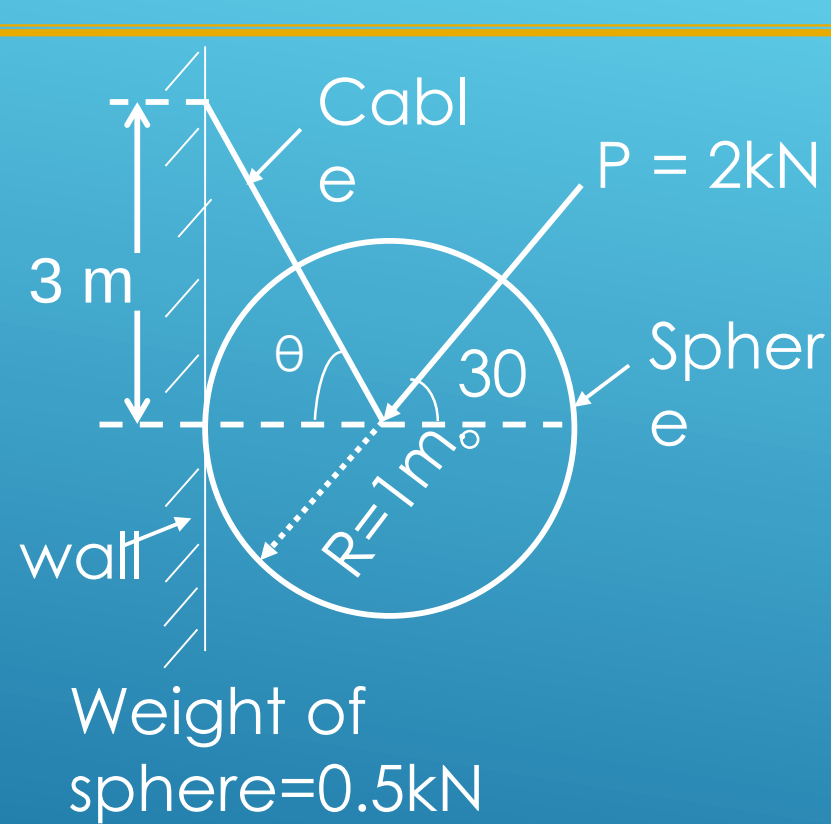
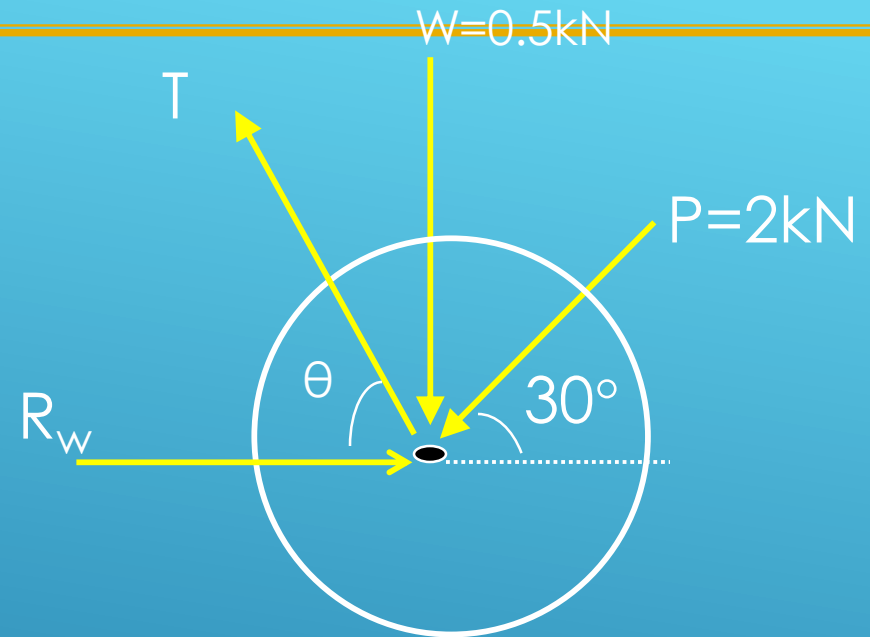


Fig 3(a) SPD



T = Tension in the cable
 R_w = Reaction of the wall

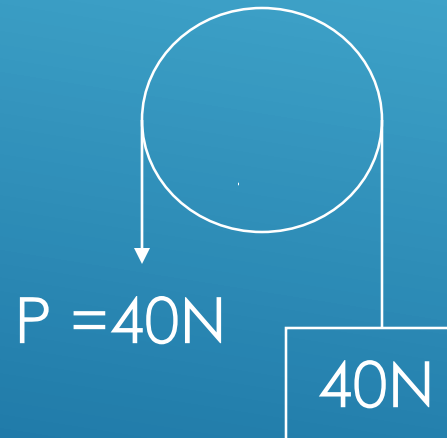
Fig 3(b) FBD

Note: Free Body Diagrams should be NEAT, LEGIBLE & SUFFICIENTLY BIG. Only the details required for the analysis of the problem are to be shown.

- 1) Tensile Force: It is a force trying to pull or extend the body. It is represented by a vector directed away from the body.
- 2) Compressive Force: It is force trying to push or contract the body. It is represented by a vector directed towards the body.
- 3) Reactions at smooth surfaces: The reactions of smooth surfaces, like walls, floors, Inclined planes, etc. will be normal to the surface and pointing towards the body.
- 4) Forces in Link rods/connecting rods: These forces will be acting along the axis of the rod, either towards or away from the body. (They are either compressive or tensile in nature).

5) Forces in Cables (Strings or Chords): These can only be tensile forces. Thus, these forces will be along the cable and directed away from the body.

6) Tension in cables on either side of a smooth pulley will be equal in magnitude. (Eg. As shown in Fig)



P (1)

NUMERICAL EXAMPLES

(1) A sphere of 100N weight is tied to a wall by a string as shown in fig (1). Find the tension in the string and the reaction of the wall.

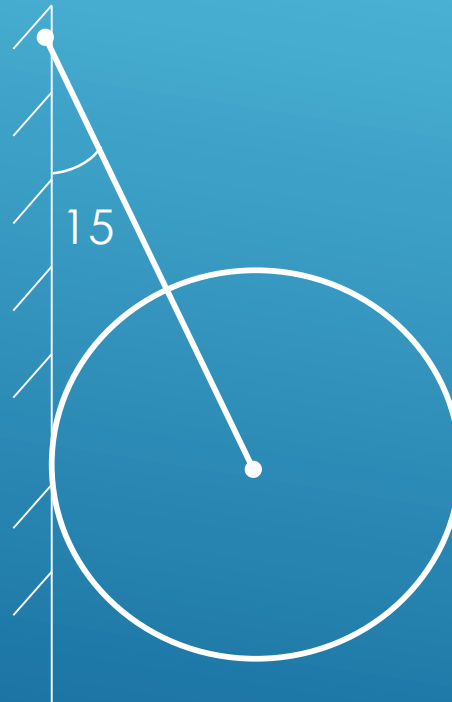


Fig (1)

P (2)

Using Lami's theorem,

$$\frac{T}{\sin 90} = \frac{W}{\sin 105} = \frac{R}{\sin 165}$$

$$T = \frac{100}{\sin 105} \times \sin 90 = 103.5 N$$

$$R = \frac{100}{\sin 90} \times \sin 165 = 26.8 N$$

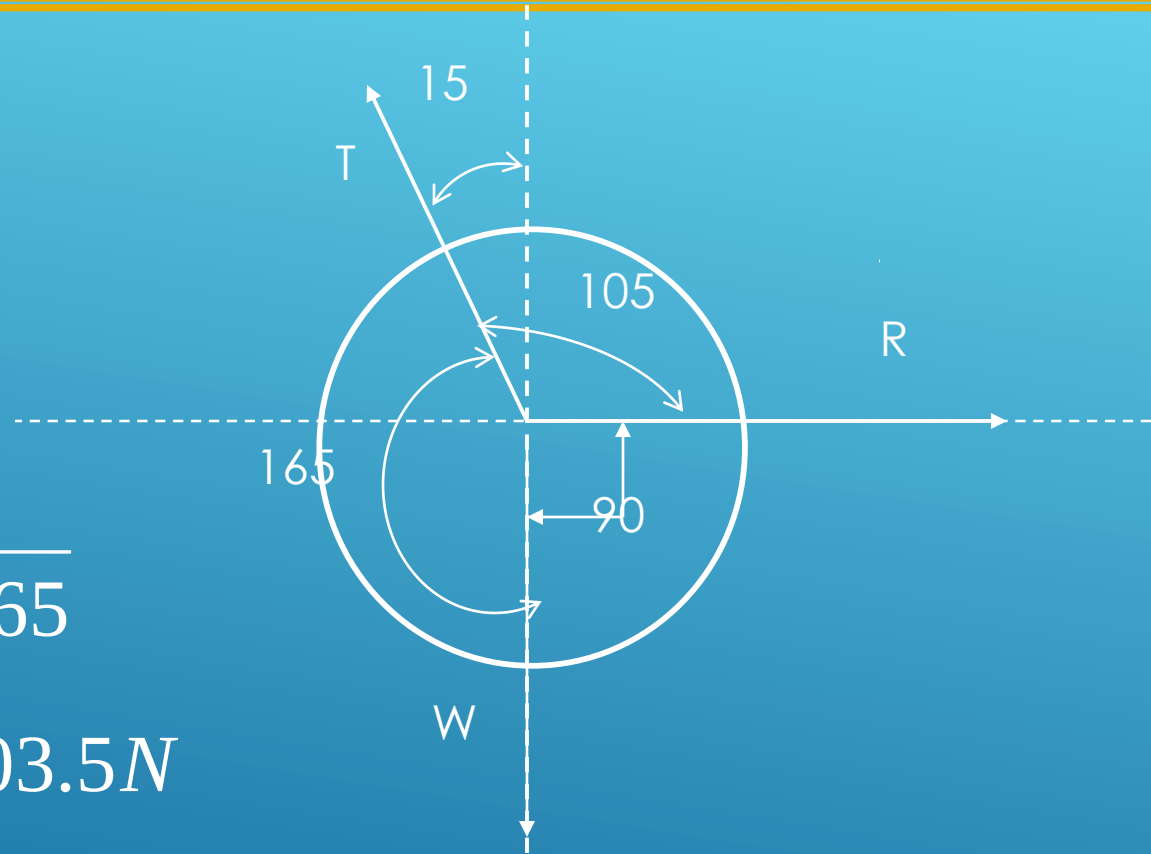


Fig (1a) FBD

P (3)

(2) Determine the magnitude and nature of the forces in the bars AB and AC shown in Fig (2). Neglect size and weight of the pulley.

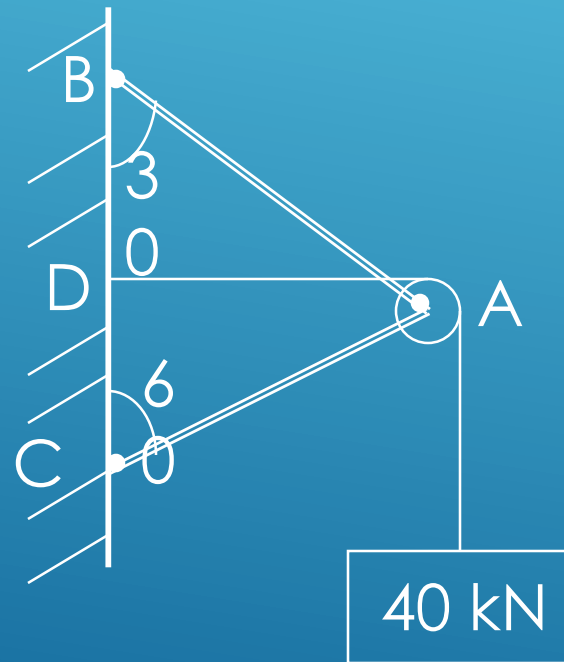


Fig (2) SPD

(P4)

Angle between F_{AB} and $F_{AC} = 90^\circ$

Taking F_{AC} as X-axis and
 F_{AB} as Y-axis

$$\begin{aligned} \text{From } +\sum F_x &= 0, \\ -F_{AC} - W_x - T_x &= 0 \\ \therefore F_{AC} &= -40\cos 60 - 40\cos 30 \\ &= -54.641\text{ kN} \end{aligned}$$

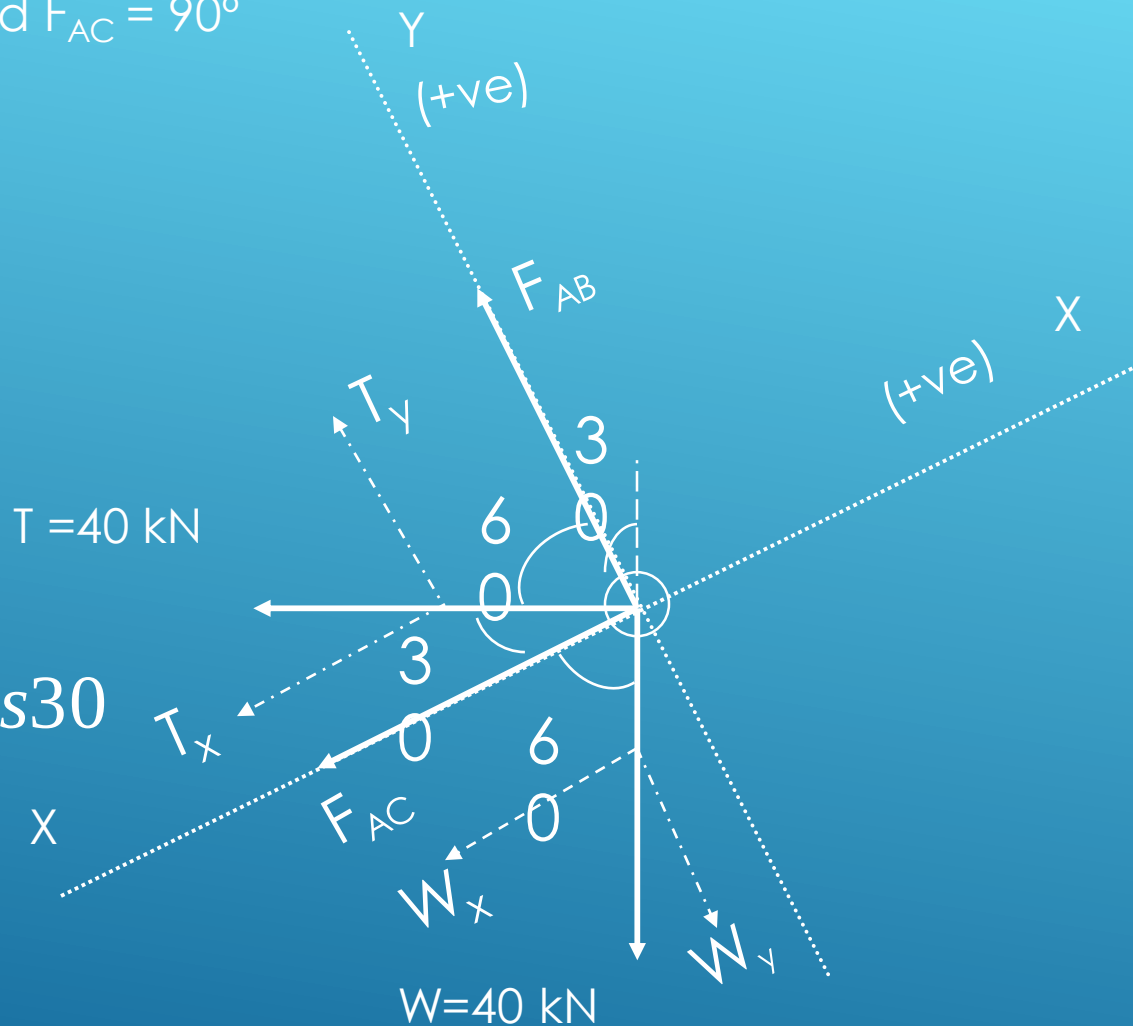


Fig (2a) F B D at A

(P5)

F_{AC} is -ve, F_{AC} is towards 'A',
So it is Compressive.

From $\uparrow \sum F_y = 0,$

$$+F_{AB} + T_Y - W_Y = 0$$

$$\therefore F_{AB} = 40 \sin 60 - 40 \sin 30$$

$$= +14.64 \text{ kN}$$

F_{AB} is +ve. F_{AB} is towards 'A', So it is
Tensile.

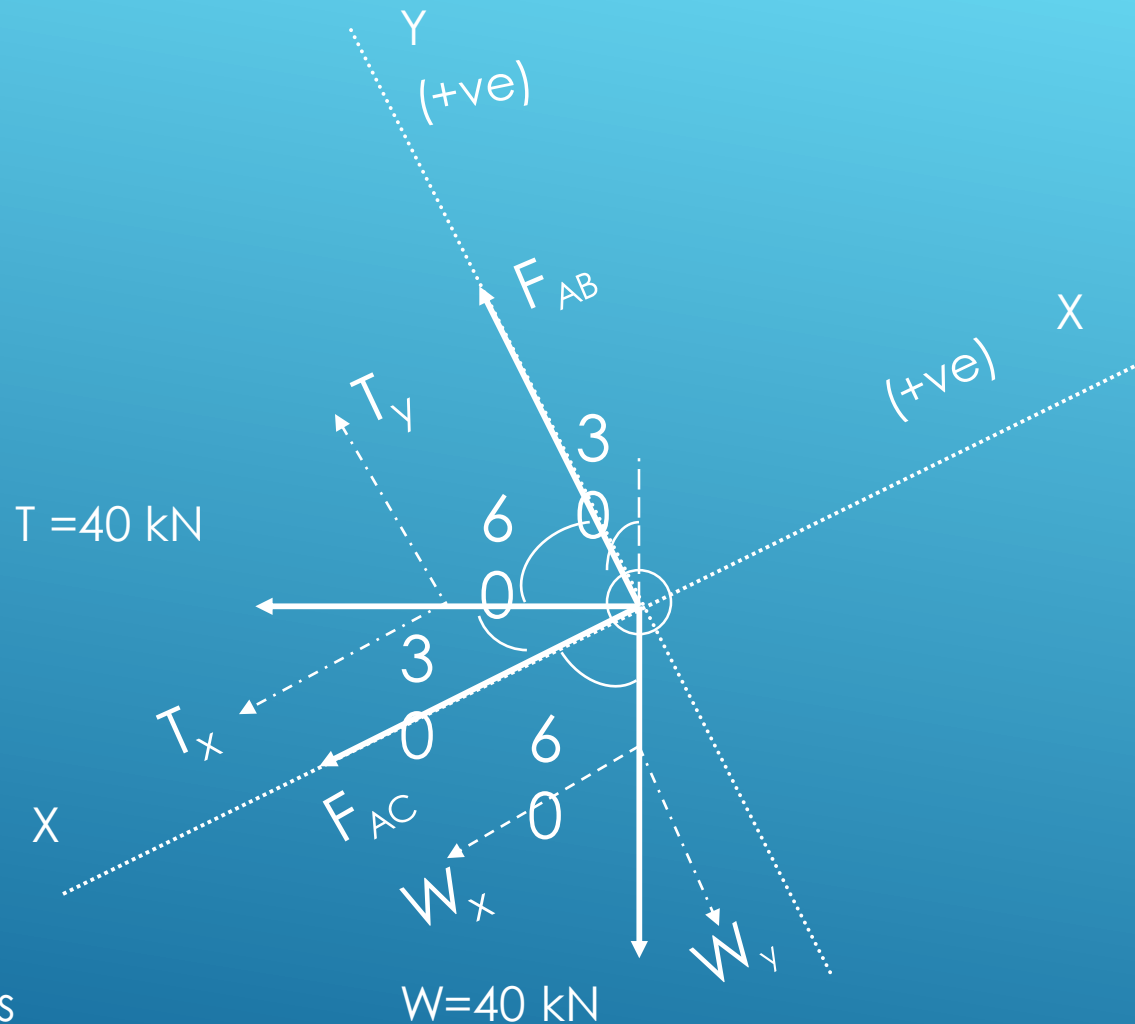


Fig (2a) F B D at A

P (6)

(3) Two cylinders A & B of weight 400N and 200N respectively, rest on smooth planes as shown in Fig(3). Find the force 'P' required for equilibrium.

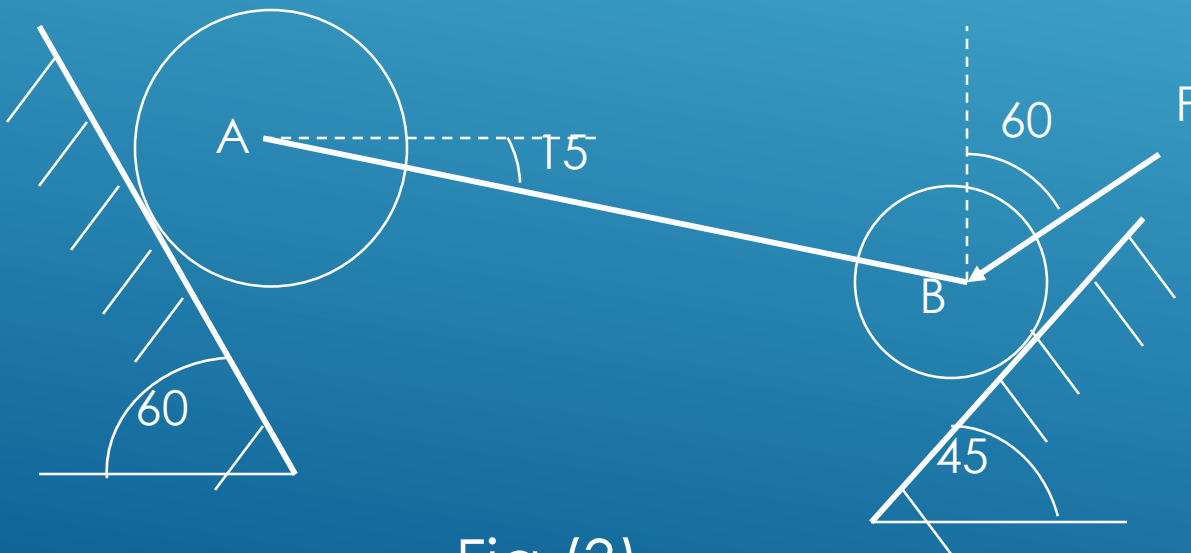


Fig (3)

P (7)

The forces in the system are as shown.

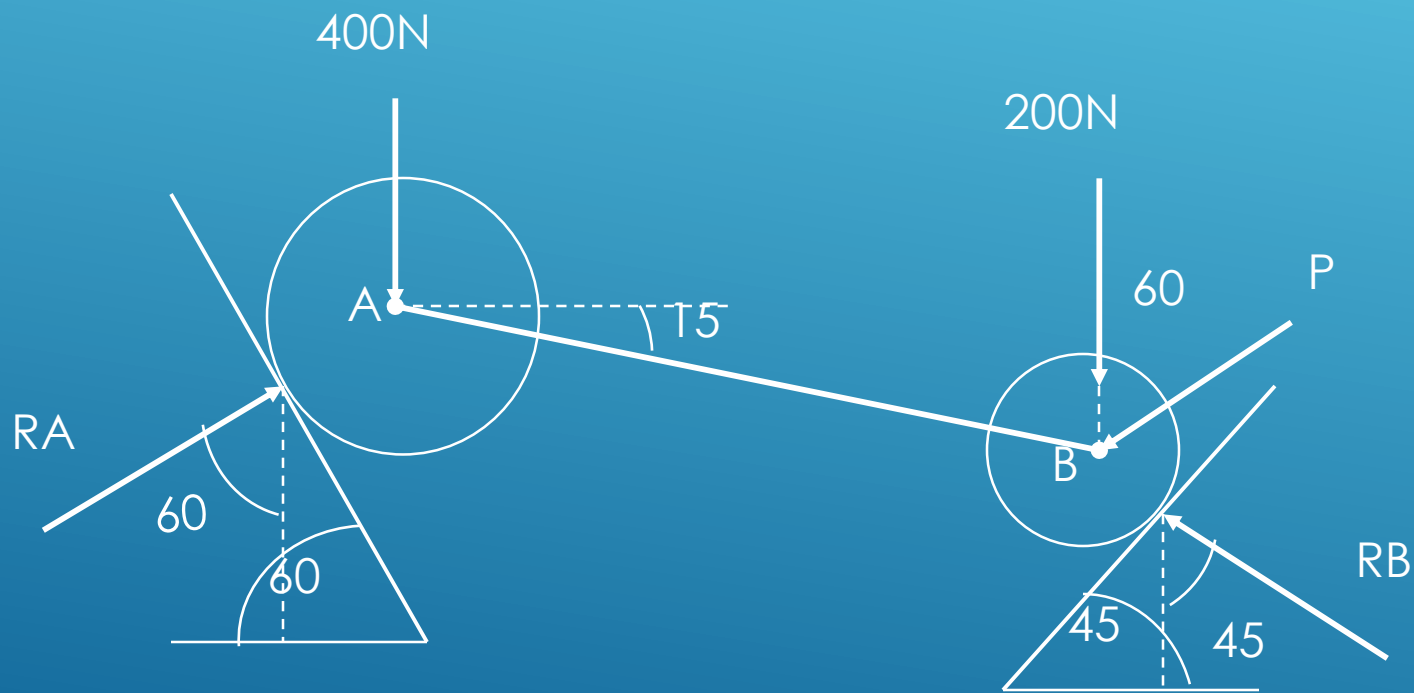


Fig (3)

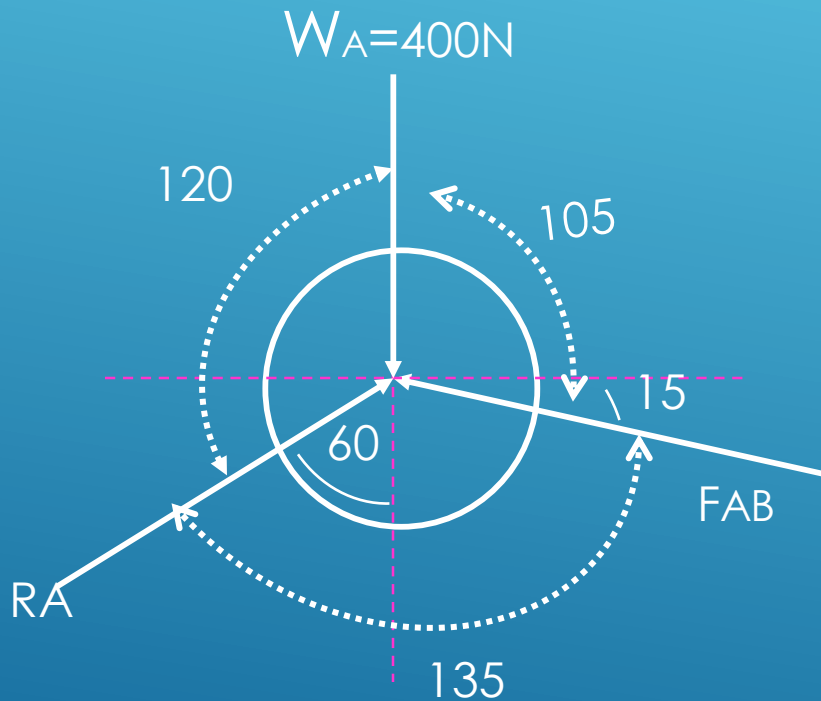


Fig (3a) FBD OF 'A'

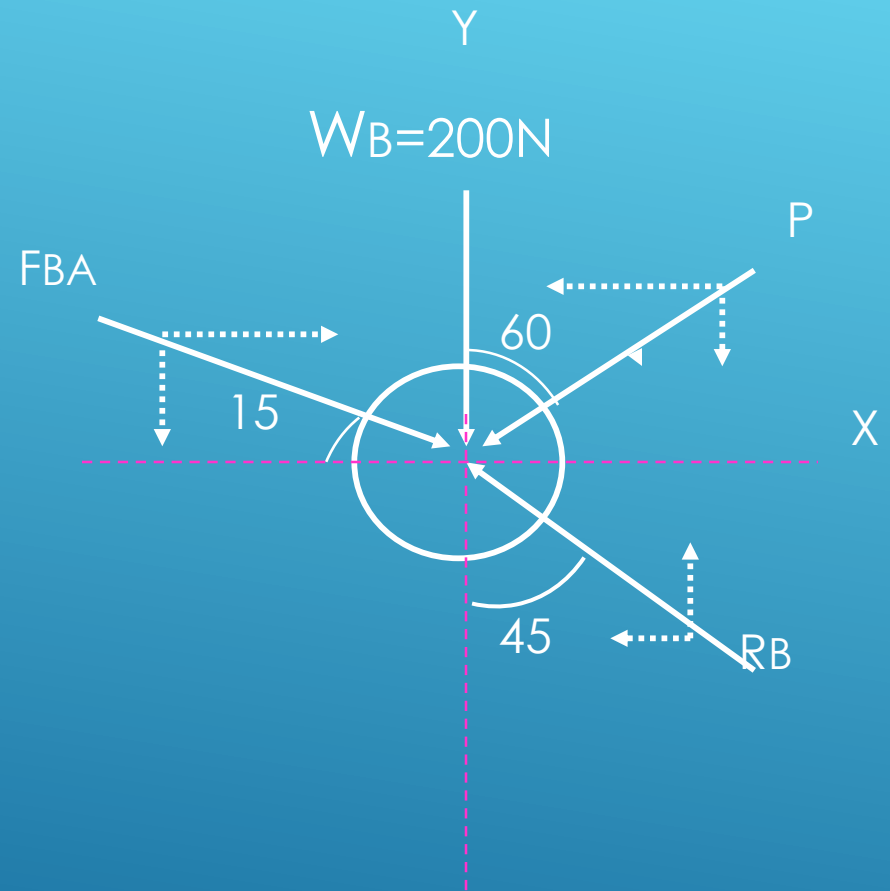


Fig (3b) FBD OF 'B'

Considering FBD of 'A' and Using Lami's theorem,

$$\frac{R_A}{\sin 105} = \frac{W_A}{\sin 135} = \frac{F_{BA}}{\sin 120}$$

$$\therefore F_{BA} = \frac{400}{\sin 135} \times \sin 120 = 489.9 \text{ N}$$

$$(R_A = \frac{400}{\sin 135} \times \sin 105 = 546.41 \text{ N})$$

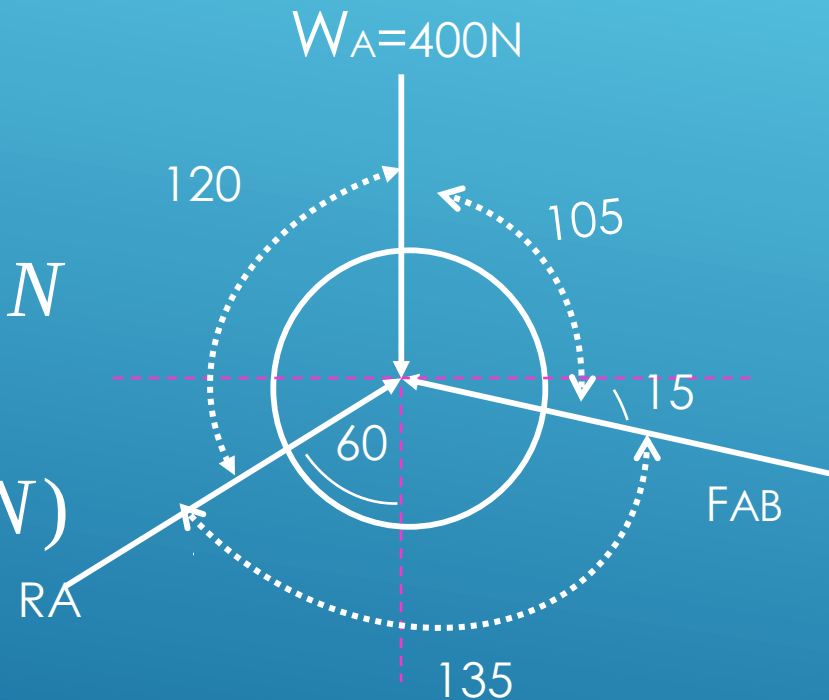


Fig (3a) FBD OF 'A'

Considering FBD of 'B', We have,

$$\text{From } \Sigma F_y = 0, \uparrow +,$$

$$-P_y + R_{BY} - W_B - F_{ABY} = 0$$

$$\therefore P \cos 60 - R_B \cos 45 = -200 - 489.9 \sin 15 \quad \text{-----Eqn(1)}$$

$$\text{From } \Sigma F_x = 0, \xrightarrow{+}$$

$$+F_{ABX} - P_x - R_{BX} = 0$$

$$\therefore P \sin 60 + R_B \sin 45 = 489.9 \cos 15 \quad \text{-----Eqn(2)}$$

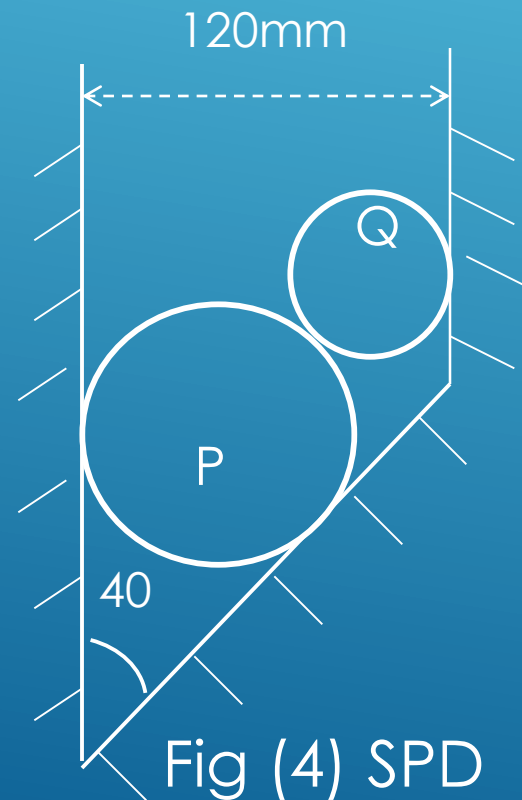
Adding Eqn(1) and Eqn(2), We get,

$$P(\cos 60 + \sin 60) = +1446.41$$

$$\therefore P = 107.18 \text{ N } (R_B = 537.94 \text{ N})$$

P (11)

(4) Two cylinders P and Q of diameters 100mm and 50mm, weighing 200N and 50N respectively, are placed in a trench as shown in Fig (4). Assuming smooth surfaces determine reactions at all contact points.



P (12)

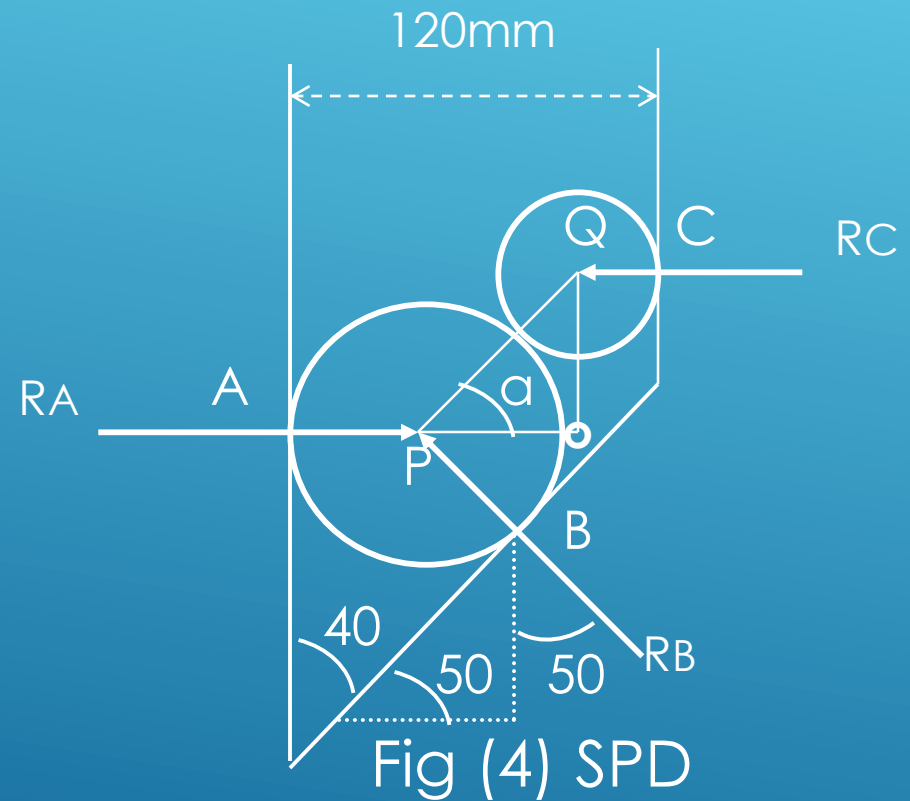
From Fig(4), in PQO

$$PQ = 50 + 25 = 75\text{mm}$$

$$PO = 120 - 50 - 25$$

PO = 45mm.

$$\cos \alpha = 45/75 ; \alpha = 53.130$$



P (13) FBD

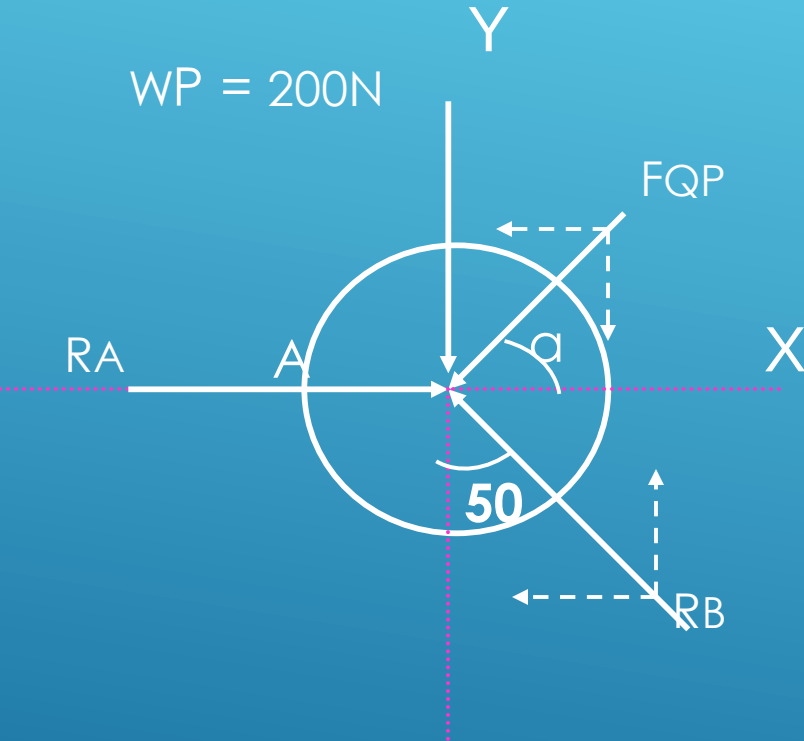


Fig (4a) FBD of 'P'

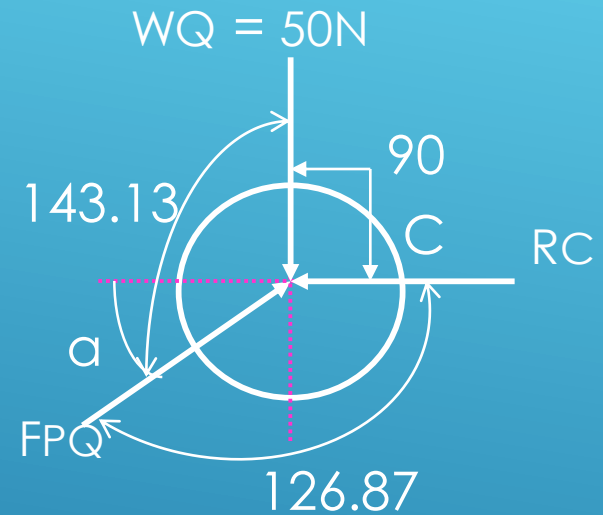


Fig (4b) FBD of 'Q'

$$\alpha = 53.13^\circ$$

Considering FBD of 'Q' and Using Lami's theorem,

$$\frac{R_C}{\sin 143.13} = \frac{W_Q}{\sin 126.87} = \frac{F_{PQ}}{\sin 90}$$

$$\therefore F_{PQ} = \frac{50}{\sin 126.87} \times \sin 90 = 62.50 \text{ N}$$

$$R_C = \frac{50}{\sin 126.87} \times \sin 143.13 = 37.50 \text{ N}$$

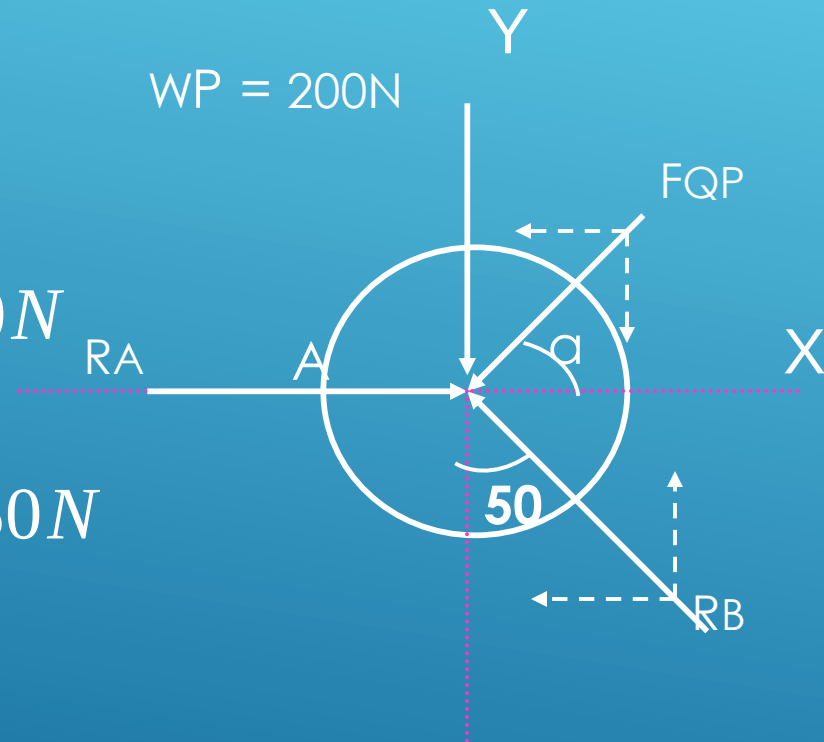


Fig (4a) FBD of 'P'

From $\Sigma F_y = 0, \uparrow +,$

$$+ R_{BY} - W_P - F_{QPY} = 0$$

$$\therefore R_B \cos 50 = 200 + 62.5 \sin 53.13$$

$$R_B = 388.93N$$

$From \sum F_x = 0, \xrightarrow{+}$

$$+R_A - F_{QPX} - R_{BX} = 0$$

$$\therefore R_A = 62.50 \cos 53.13 + 388.93 \sin 50$$

$$R_A = 335.44N$$

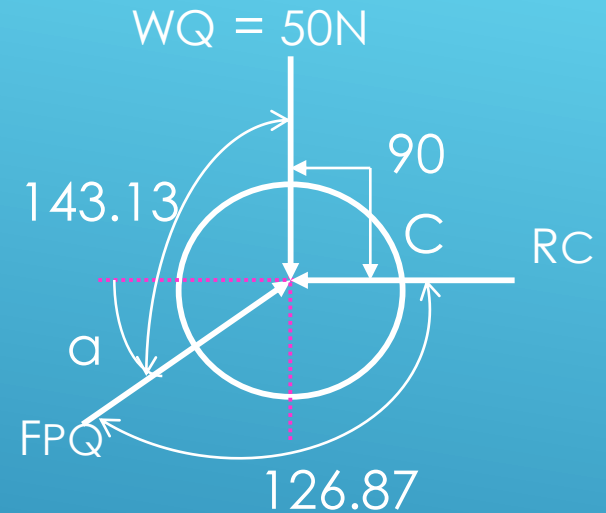


Fig (4b) FBD of 'Q'

(5) Three cylinders A, B and C of diameters 500mm each are arranged as shown in Fig (5). The weights of A and B are 500N each and weight of C is 600N. Determine reactions at all contact points and tension in the string holding A and B.

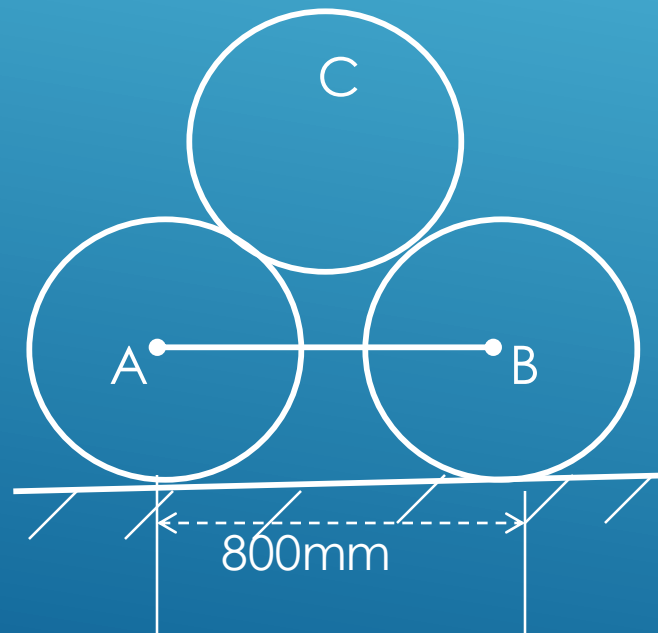


Fig (5) SPD

P (17)

From Fig(5a),

$$AC = 250 + 250 = 500\text{mm}$$

$$AO = 800/2 = 400\text{mm}$$

$$\cos \alpha = 400/500 ; \alpha = 36.87^\circ$$

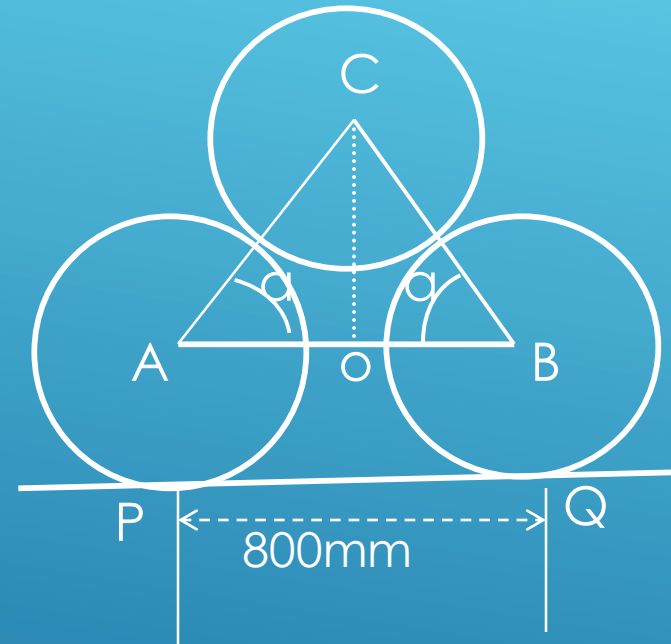


Fig (5a)
SPD

P (18) FBD

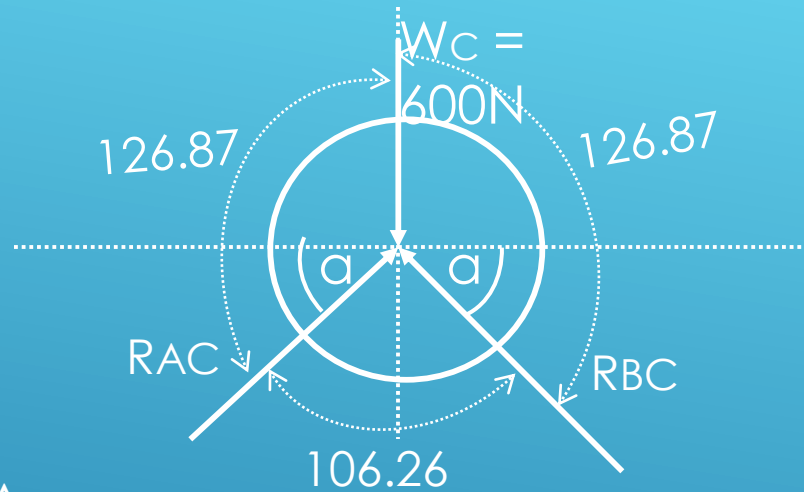


Fig (5c) FBD of 'C'

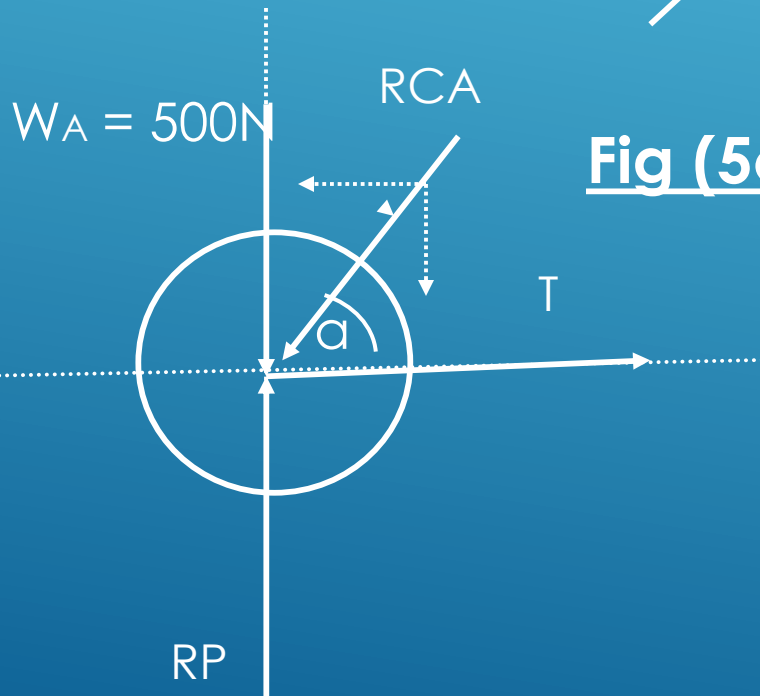


Fig (5a) FBD of 'A'

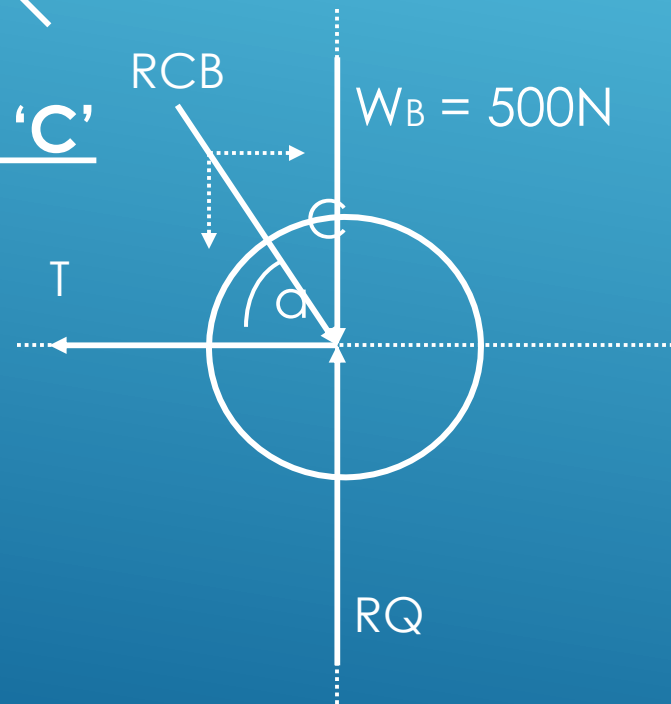


Fig (5b) FBD of 'B'

Considering FBD of 'C' and Using Lami's theorem,

$$\frac{R_{AC}}{\sin 126.87} = \frac{W_C}{\sin 106.26} = \frac{R_{BC}}{\sin 126.87}$$

$$\therefore R_{AC} = R_{BC} = \frac{600}{\sin 106.26} \times \sin 126.87 = 500N$$

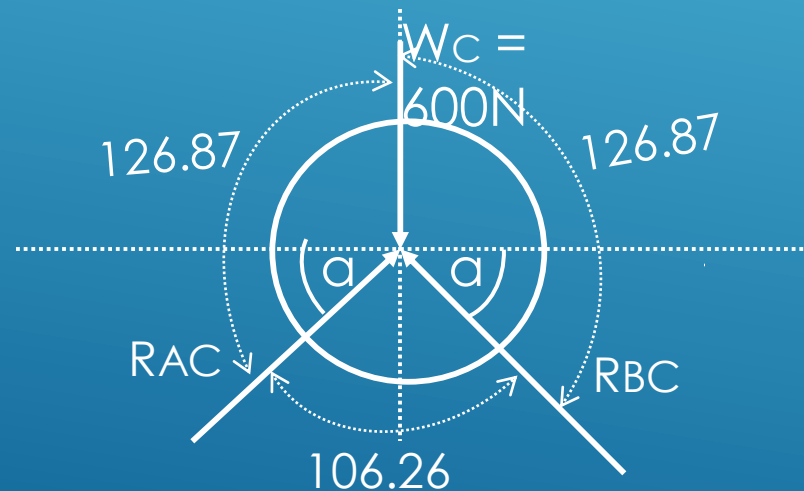


Fig (5c) FBD of 'C'

Considering FBD of 'A', we have,

$$\text{From } \sum F_x = 0, \xrightarrow{+}$$

$$+T - R_{CAx} = 0$$

$$T = 500 \cos 36.87$$

$$\therefore T = 400 \text{ N}$$

$$\text{From } \sum F_y = 0, \uparrow +,$$

$$+R_p - W_A - R_{CAy} = 0$$

$$R_p = 500 + 500 \sin 36.87$$

$$\therefore R_p = 800 \text{ N}$$

By Symmetry, $RP = RQ = 800 \text{ N}$

(Or, Using FBD of 'B', $RQ = 800 \text{ N}$)

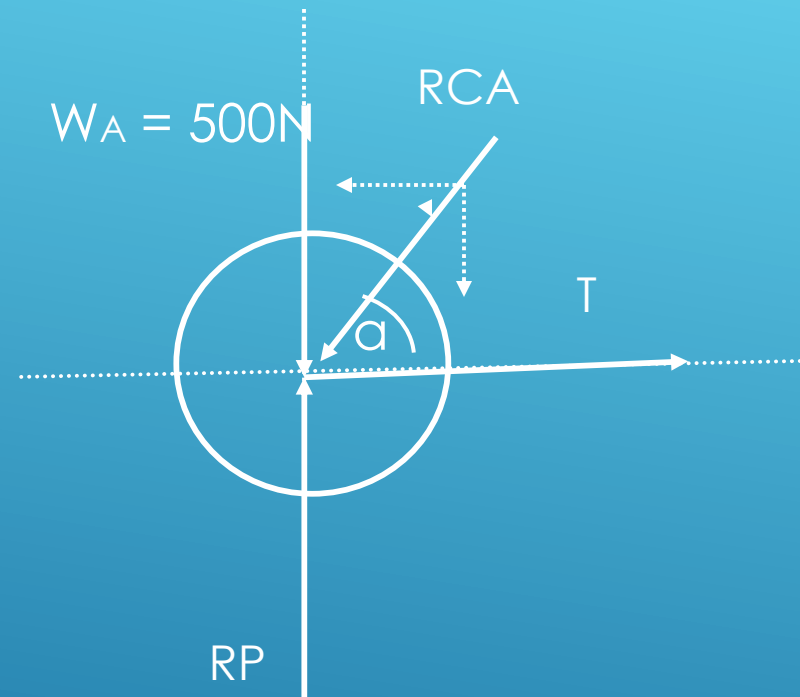


Fig (5a) FBD of 'A'

P (21)

(6) Two Spheres, each of radius 1 m and mass 1000 kg, rest on smooth surfaces as shown in Fig (6) . Determine reactions at all contact points.

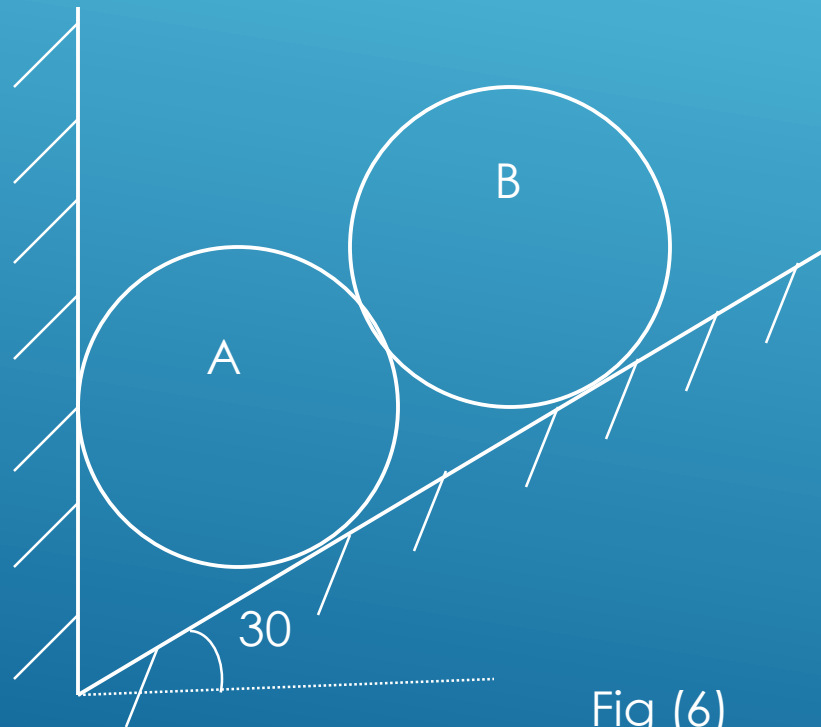
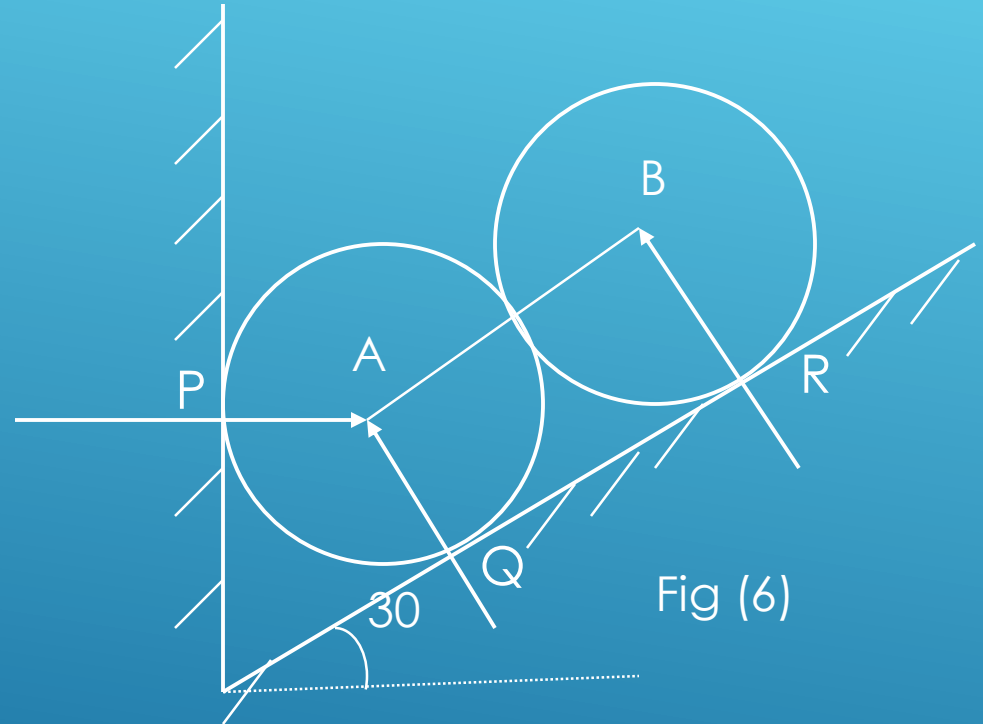


Fig (6)

Weight of Spheres
= Mass $\times$ g
= 1000×9.81
= $9810 \text{ N} = 9.81 \text{ kN}$



Reaction between A & B will be parallel to Plane QR, as radius is the same.

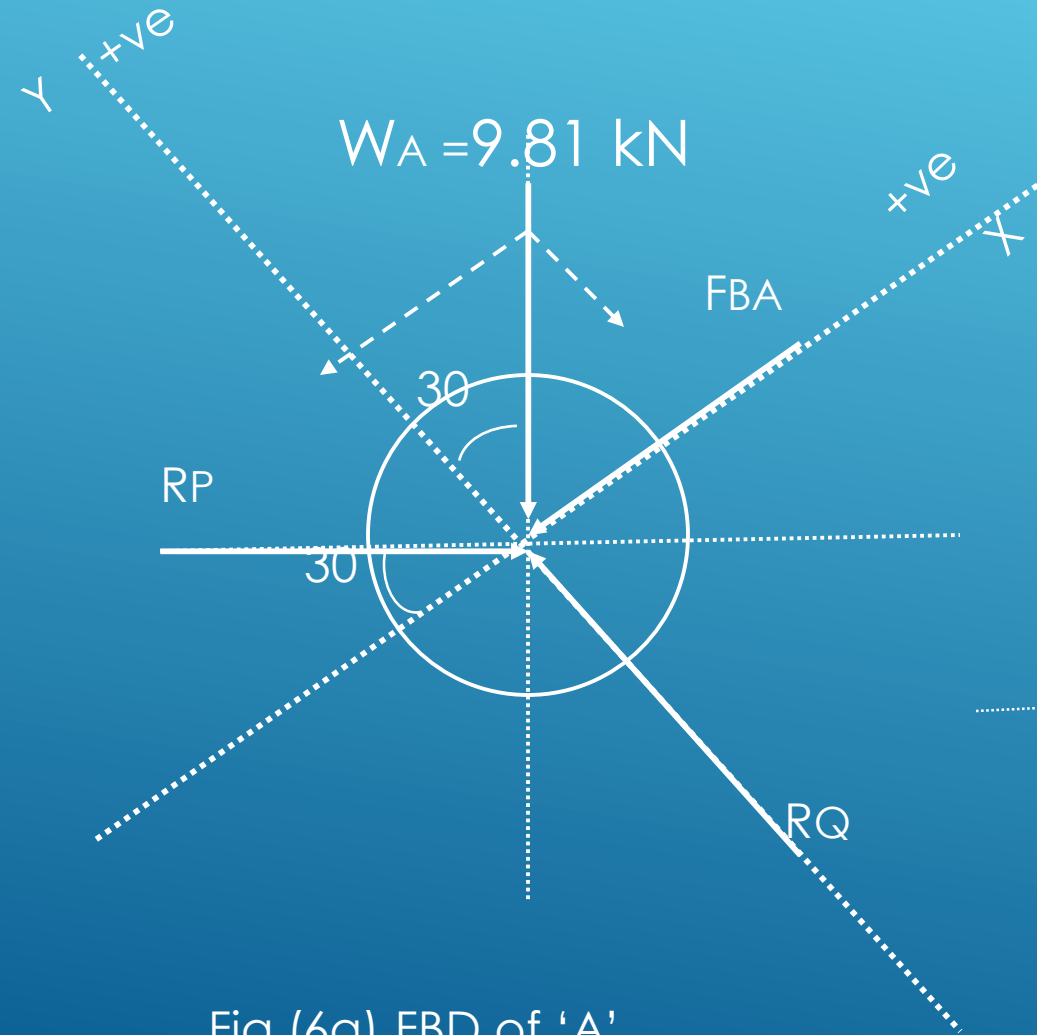


Fig (6a) FBD of 'A'

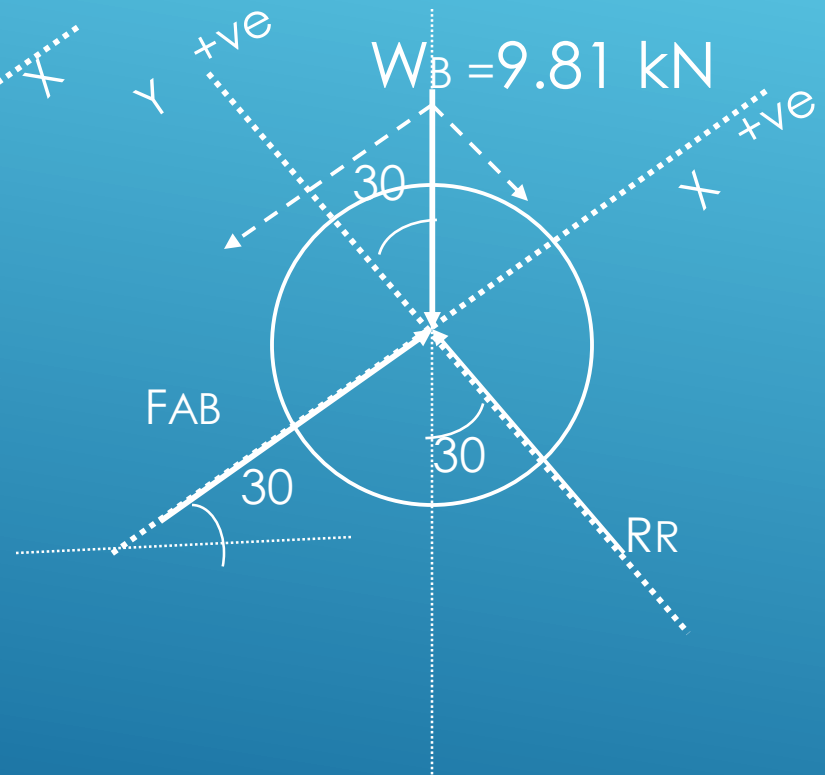


Fig (6b) FBD of 'B'

Considering FBD of 'B', We have,

$$\text{From } \Sigma F_x = 0, \nearrow +$$

$$F_{AB} = W_x = 9.81 \sin 30$$

$$\therefore F_{AB} = 4.9 \text{ kN}$$

$$\text{From } \Sigma F_y = 0, \nwarrow +$$

$$R_R = W_y = 9.81 \cos 30$$

$$\therefore R_R = 8.49 \text{ kN}$$

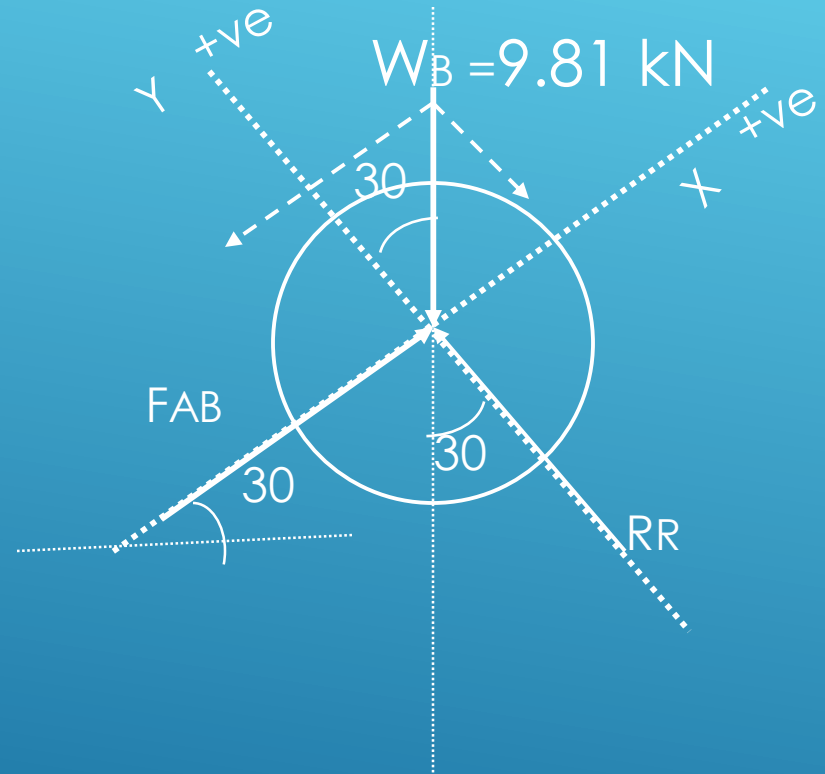


Fig (6b) FBD of 'B'

Considering FBD of 'A', we have,

$$\text{From } \sum F_x = 0, \quad \nearrow^+$$

$$R_{PX} = W_{AX} + F_{BA}$$

$$R_P \cos 30 = 9.81 \sin 30 + 4.9$$

$$\therefore R_P = 11.32 \text{ kN}$$

$$\text{From } \sum F_y = 0, \quad \nwarrow^+$$

$$R_Q = W_{AY} + R_{PY}$$

$$R_Q = 9.81 \cos 30 + 11.32 \sin 30$$

$$\therefore R_Q = 14.145 \text{ kN}$$

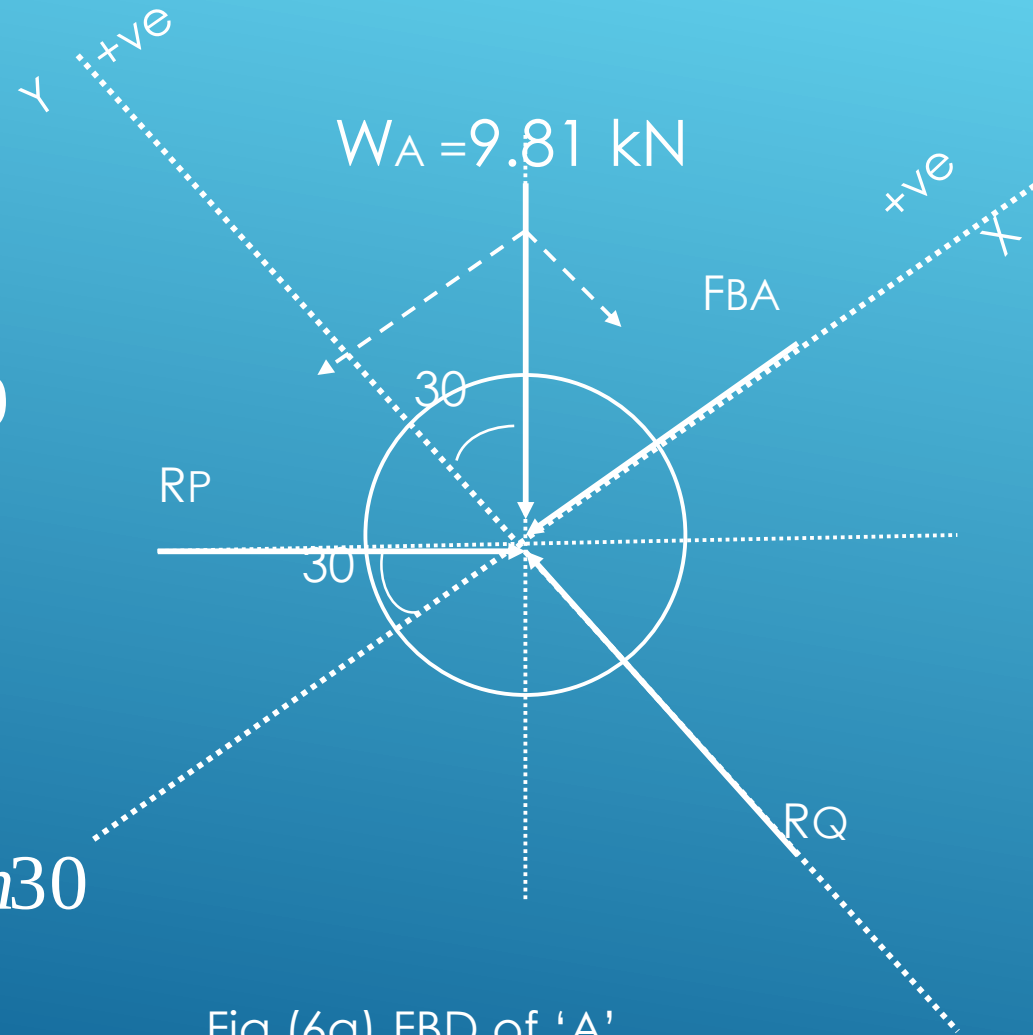
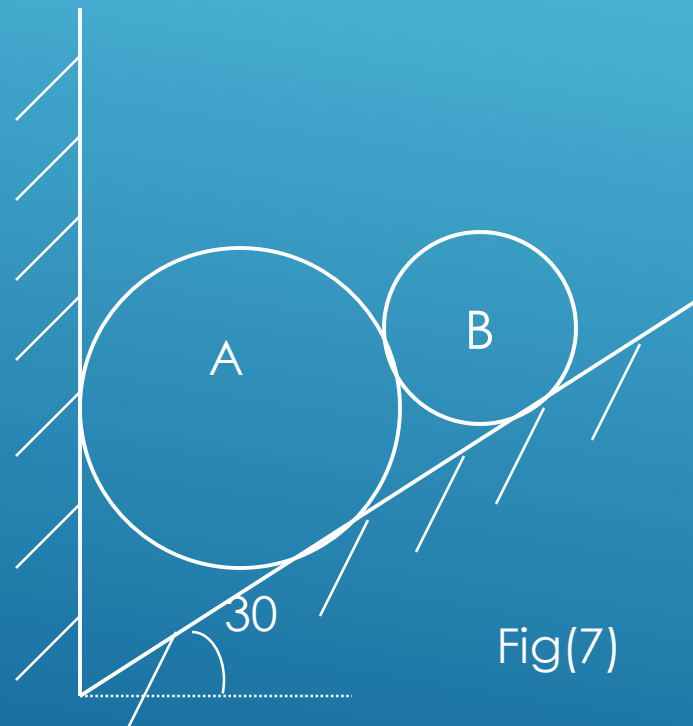


Fig (6a) FBD of 'A'

(7) Two Spheres A & B, weighing 300N & 600N and having diameters 800mm & 1200mm, respectively, rest on smooth surfaces as shown in Fig (7) . Determine reactions at all contact points.



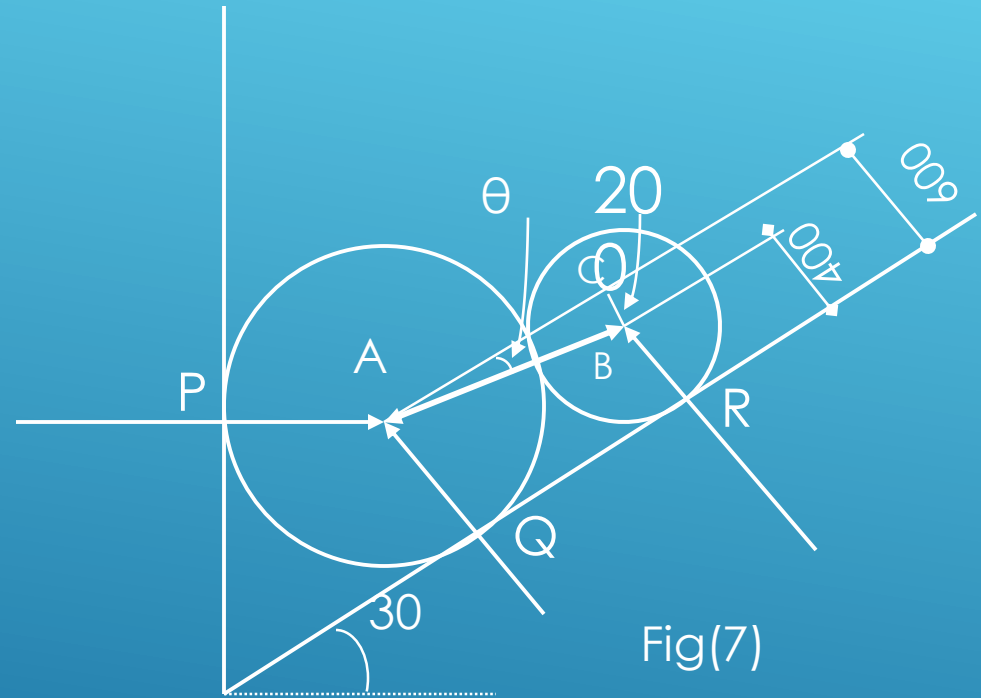
Fig(7)

AC parallel to plane;
AB radial line.

$$\sin \theta = BC/AB$$

$$\sin \theta = 200/1000 = 11.53^\circ$$

Force between A & B
will be at an angle of 11.53°
to axis parallel to Plane QR.



(Radii are NOT the same in this case.)

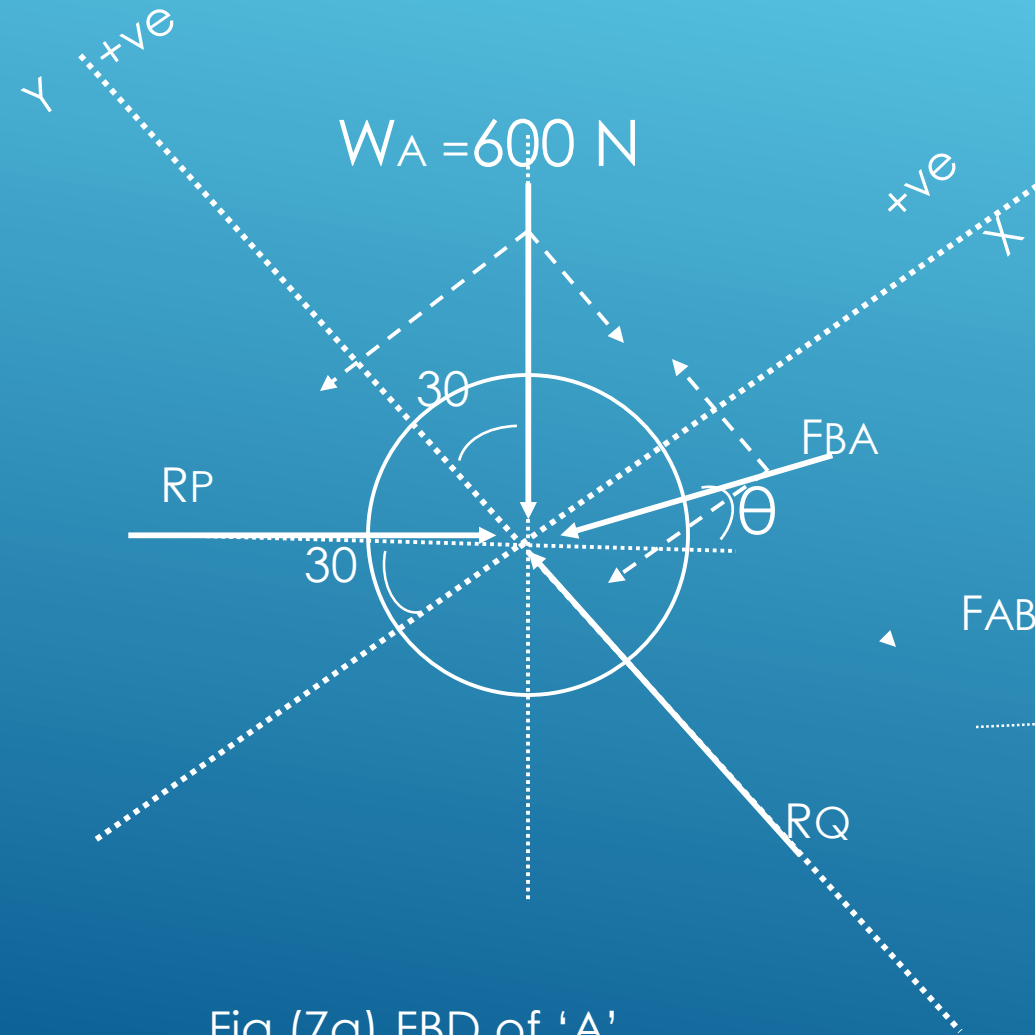


Fig (7a) FBD of 'A'

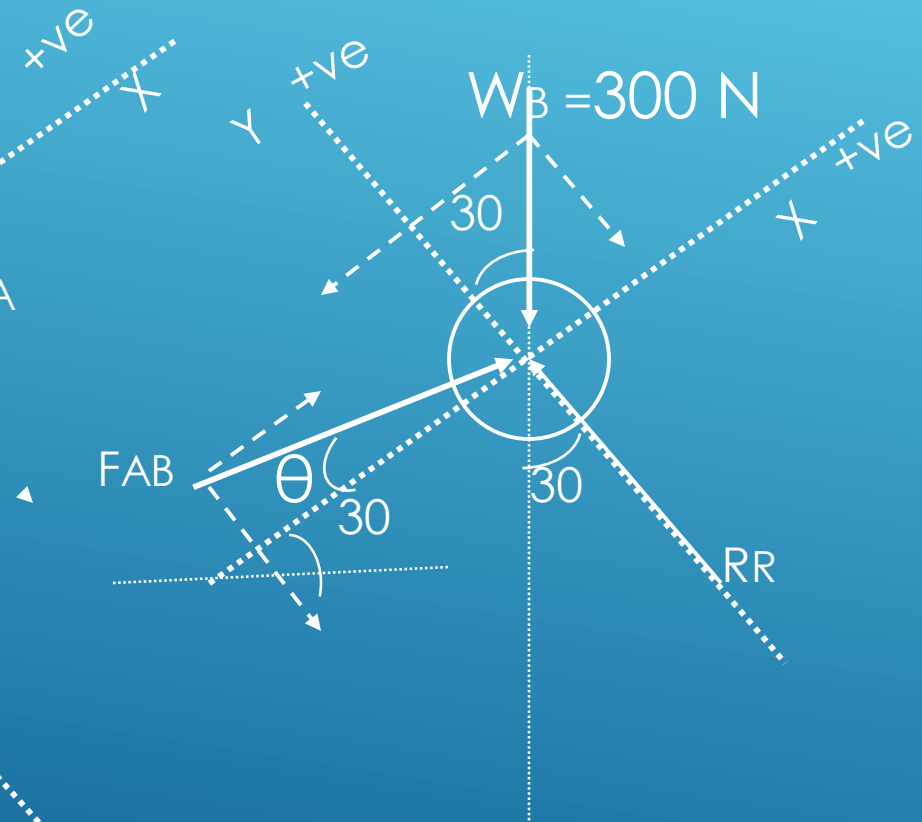


Fig (7b) FBD of 'B'

Considering FBD of 'B', We have,

$$\text{From } \Sigma F_x = 0,$$

$$F_{ABX} = W_{AX}$$

$$\therefore F_{AB} = \frac{300 \sin 30}{\cos 11.53} = 153.09 \text{ N}$$

$$\text{From } \Sigma F_y = 0,$$

$$R_R = W_{AY} + F_{ABY}$$

$$R_R = 300 \cos 30 + 153.09 \sin 11.53$$

$$\therefore R_R = 290.43 \text{ N}$$

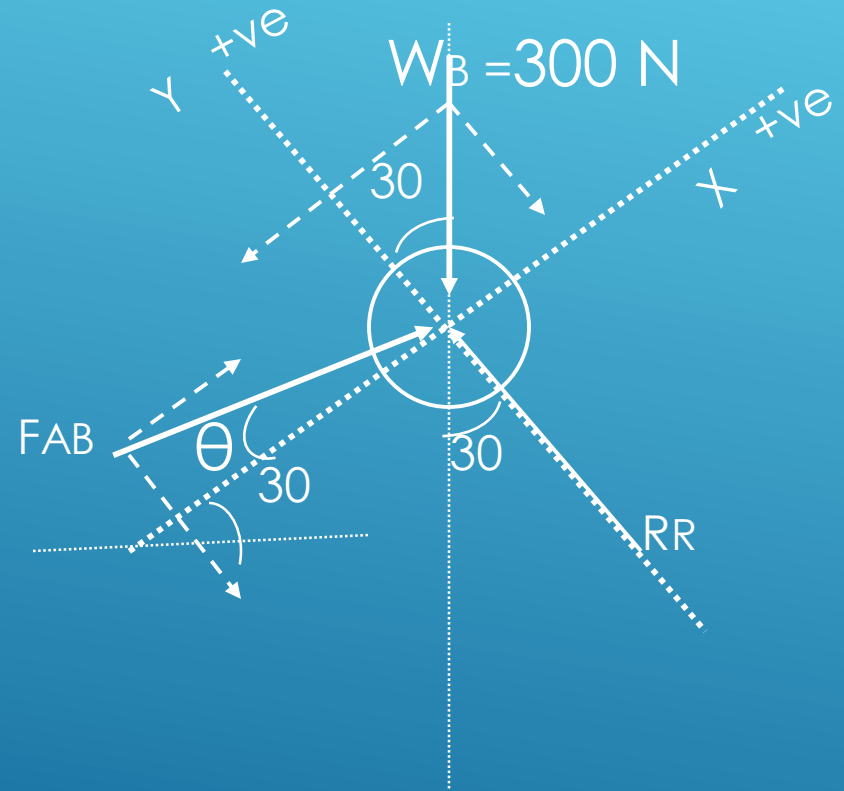


Fig (7b) FBD of 'B'

Considering FBD of 'A', We have,

$$\text{From } \Sigma F_x = 0, \quad + \nearrow$$

$$R_{PX} = W_{AX} + F_{BAX}$$

$$R_P \cos 30 = 600 \sin 30 + 153.09 \cos 11.53$$

$$\therefore R_P = 519.61 \text{ N}$$

$$\text{From } \Sigma F_y = 0, \quad \nwarrow +$$

$$R_Q = W_{AY} + R_{PY} - F_{BAY}$$

$$R_Q = 600 \cos 30 + 519.61 \sin 30 - 153.09 \sin 11.53$$

$$\therefore R_Q = 748.80 \text{ N}$$

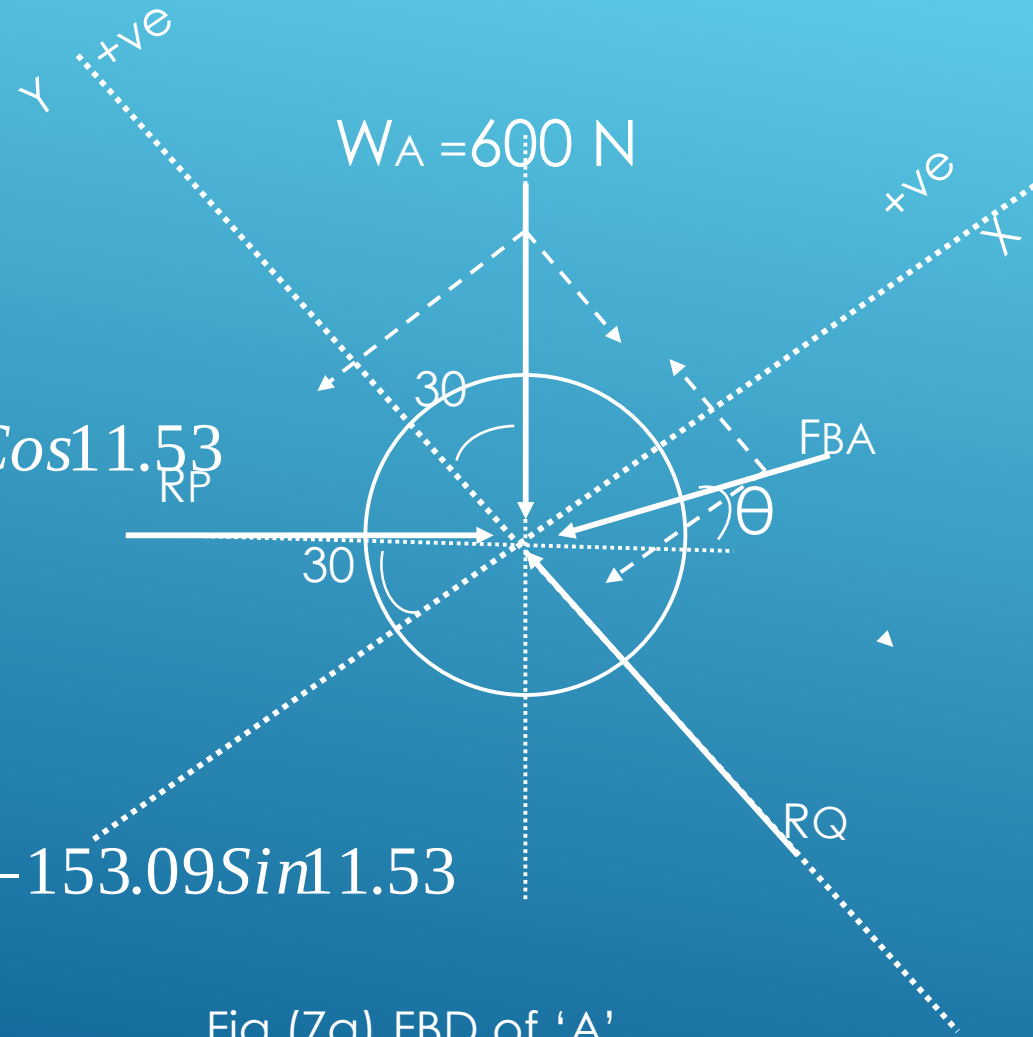
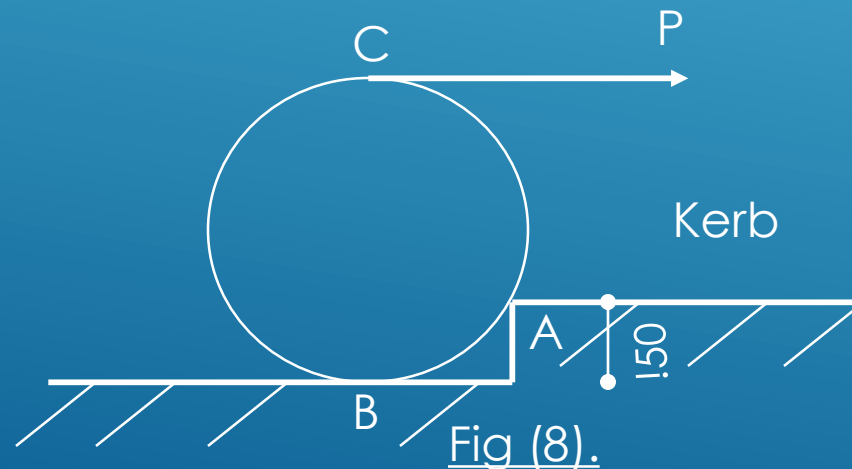


Fig (7a) FBD of 'A'

(8) A roller of radius 300mm, weighing 5kN is to be pulled over a kerb of height 150mm, by applying a horizontal force 'P' applied at the circumference by means of a rope wrapped around it, as shown in Fig (8). Find (i) The magnitude of force P when it is horizontal ; (ii) The direction and magnitude of the least force P required to pull the roller over the kerb.

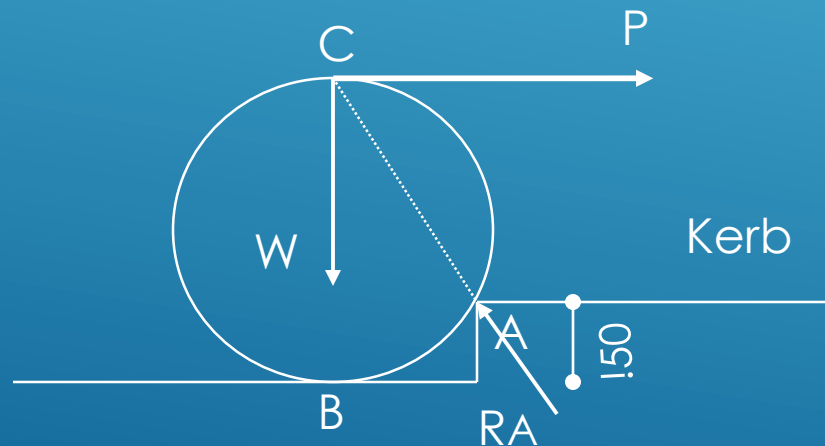


NOTE: When the roller is about to roll over the kerb, it loses contact at B. So, there will be no reaction at B. There will be a reaction at A, (say, R_A), only.

Therefore at the instant of rolling over the kerb, there will be only 3 forces in equilibrium, viz., Weight (W), P & R_A .

Since, there are only 3 forces in equilibrium, **they will be Concurrent.**

Thus reaction at A passes through 'C', point of concurrence of W & P .



In Fig (8), From Triangle ADO,

$$AD = \sqrt{AO^2 - OD^2}$$

$$\therefore AD = \sqrt{300^2 - 150^2} = 259.80mm$$

From Triangle ACD,

$$\tan \theta = \frac{CD}{AD} = \frac{450}{259.80}$$

$$\therefore \theta = 60^\circ$$

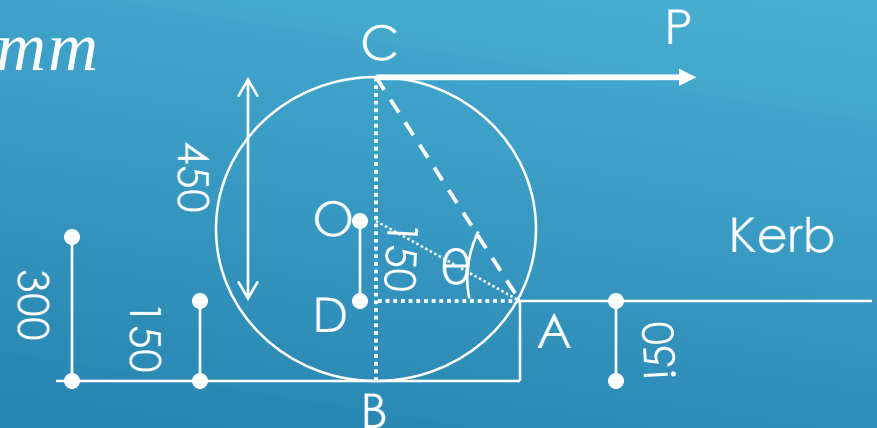


Fig (8)

CASE (i) Force 'P' is Horizontal

In Fig (8A), FBD, using Lami's Theorem,

$$\theta = 60^\circ$$

$$\frac{P}{\sin 150} = \frac{W}{\sin 120}$$

$$P = \frac{5}{\sin 120} \times \sin 150$$

$$\therefore P = 2.89 \text{ kN}$$

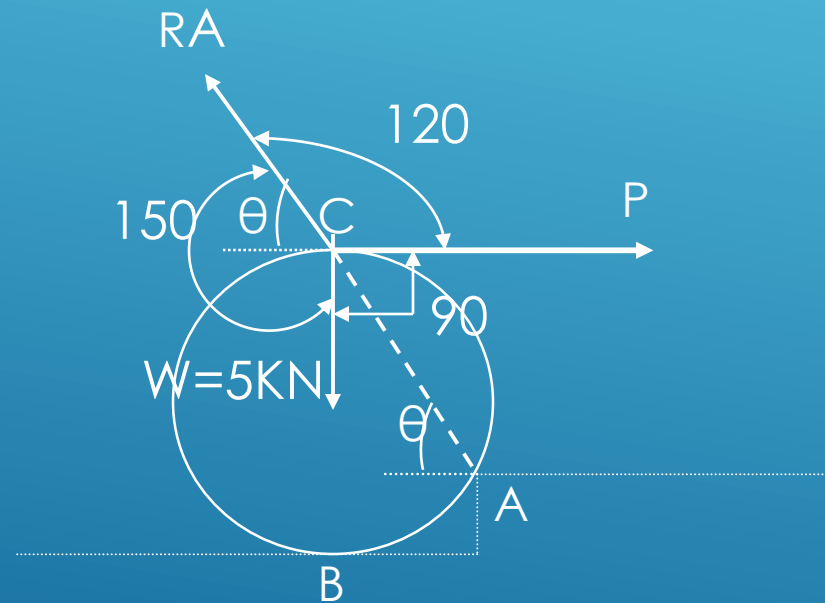


Fig (8A) FBD

CASE (ii) Force 'P' is Least. (at angle 'α' w r t Hz.)

In Fig (8B), FBD, using Lami's Theorem,

$$\theta = 60^\circ$$

$$\frac{P_{\min}}{\sin 150} = \frac{W}{\sin(120 - \alpha)}$$

$$\therefore P_{\min} = \frac{5}{\sin(120 - \alpha)} \times \sin 150$$

For P to be min., $\sin(120 - \alpha) = 1$,

$$P_{\min} = \frac{5}{\sin 90} \times \sin 150 = 2.5 \text{ kN}$$

or, $\alpha = 30^\circ$ Thus,

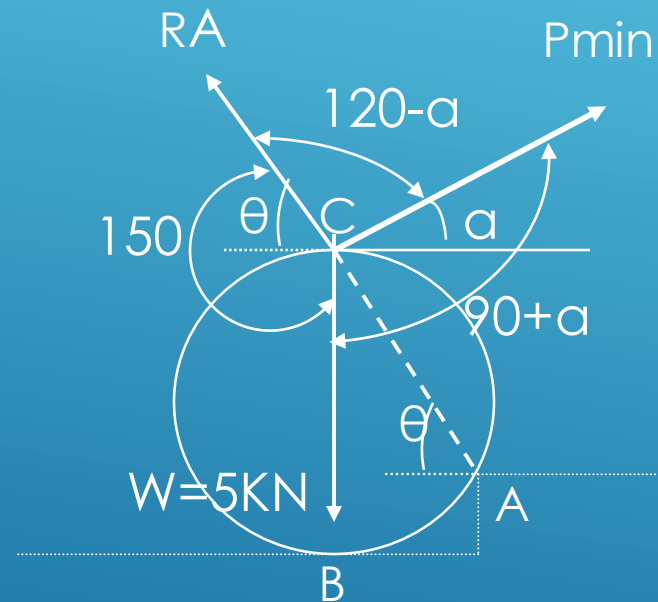


Fig (8B) FBD Case ii

NOTE: For min. value P is at right angles to R_A .

(9) Determine, the tension in the strings AB, BC, CD and inclination of the segment CD to the vertical, in the system shown in Fig (9).

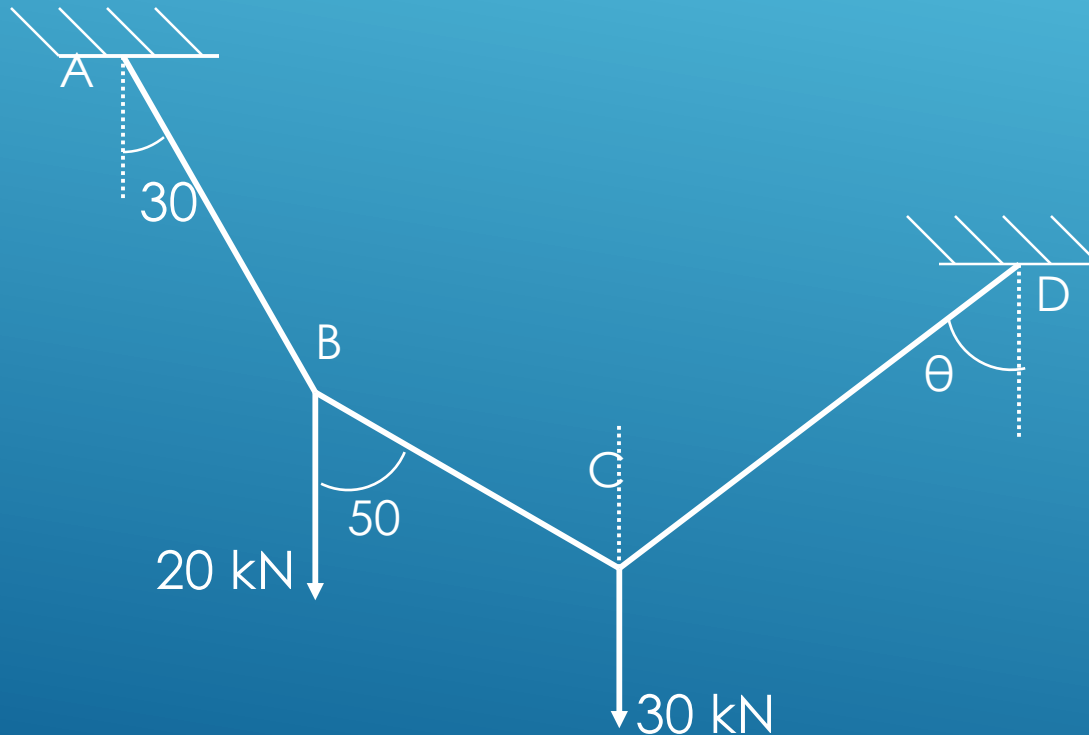


Fig (9) SPD

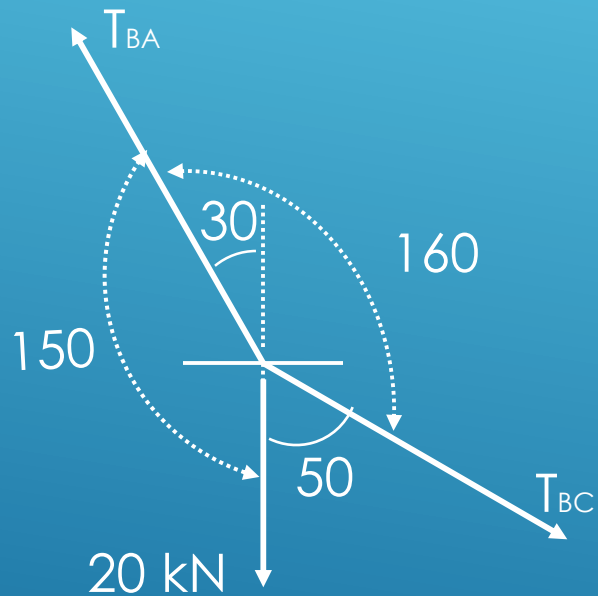


Fig 9 (a) FBD of Joint 'B'

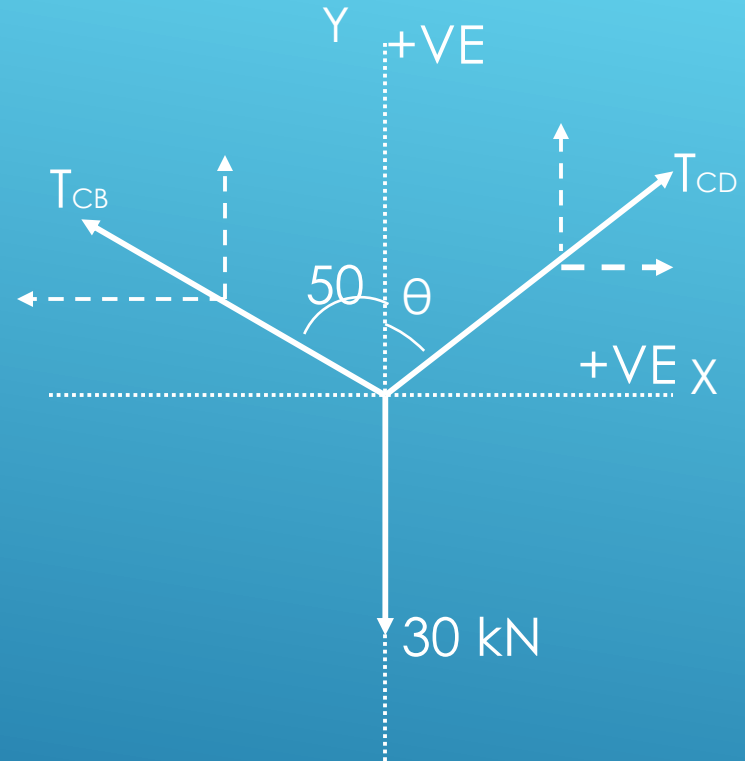
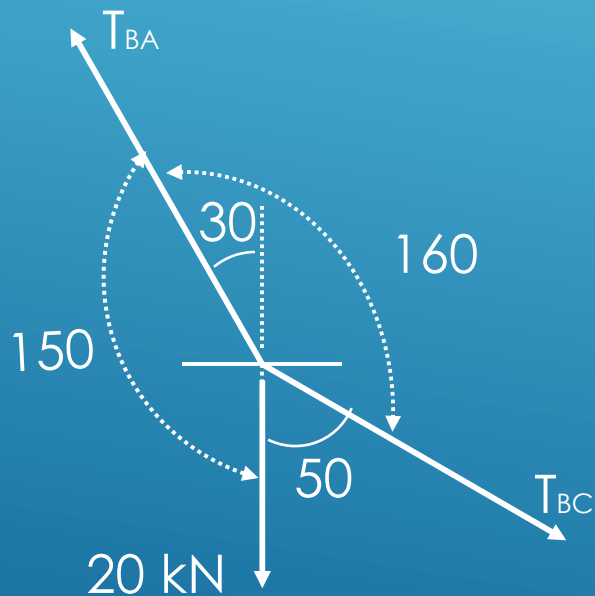


Fig 9 (b) FBD of Joint 'C'

Considering FBD of Joint 'B' and Using Lami's theorem,



$$\frac{T_{BA}}{\sin 50} = \frac{20}{\sin 160} = \frac{T_{BC}}{\sin 150}$$

$$\therefore T_{BA} = \frac{20}{\sin 160} \times \sin 50 = 44.80 \text{ kN}$$

$$T_{BC} = \frac{20}{\sin 160} \times \sin 150 = 29.24 \text{ kN}$$

Fig 9 (a) FBD of Joint 'B'

Considering FBD of Joint 'C', We have,

$$\text{From } \sum F_x = 0, \longrightarrow$$

$$T_{CD} \sin \theta = 29.24 \sin 50$$

$$\text{From } \sum F_y = 0, + \uparrow$$

$$T_{CD} \cos \theta = 30 - 29.24 \cos 50$$

Dividing Eqn(1) by (2), we get,

$$\tan \theta = \frac{22.40}{11.20}$$

$$\theta = 63.42^\circ$$

$$\therefore T_{CD} = 25.05 \text{ kN}$$

(NOTE: For this FBD, if we use Lami's Theorem, we have to expand $\sin(50+\theta)$ and solve for θ , which can take more time.)

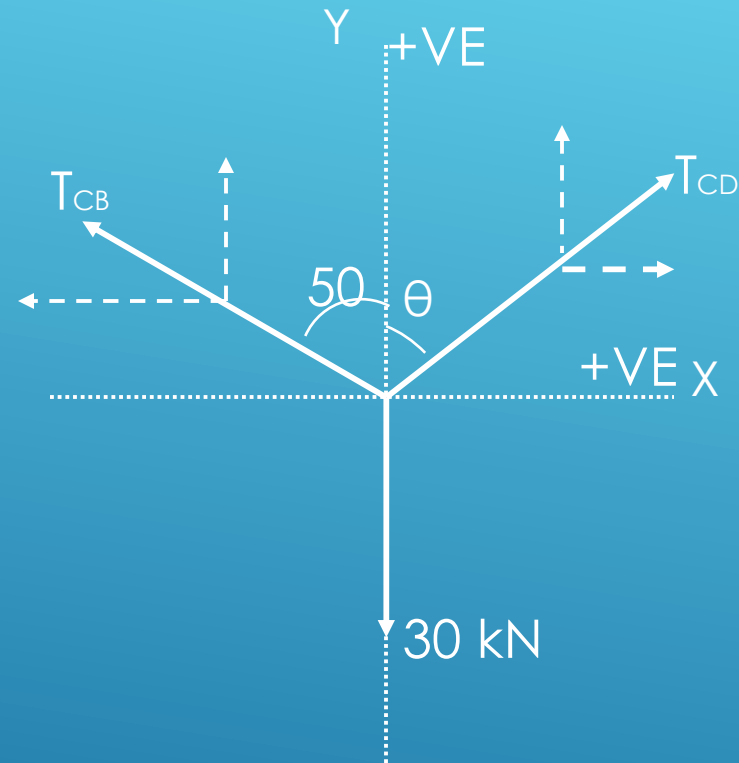


Fig 9 (b) FBD of Joint 'C'

(10) A wire is fixed at two points A and D as shown in Fig (10). Determine inclination of the segment BC to the vertical and the tension in all the segments.

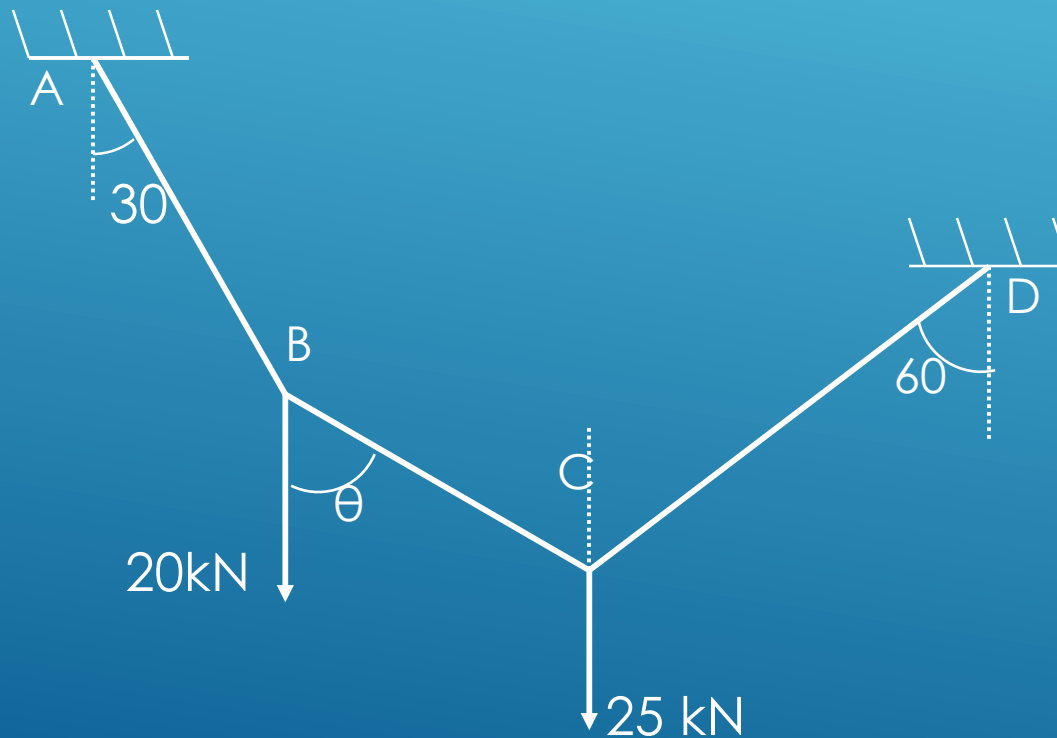


Fig (10) SPD

P (41) FBD

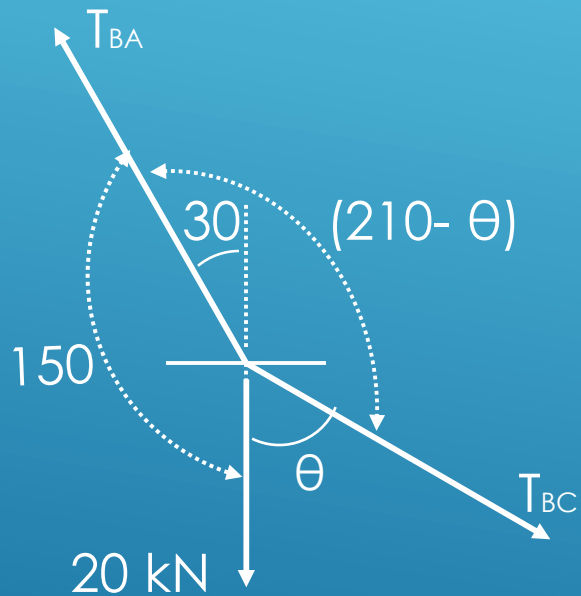


Fig (10a) FBD at Joint 'B'

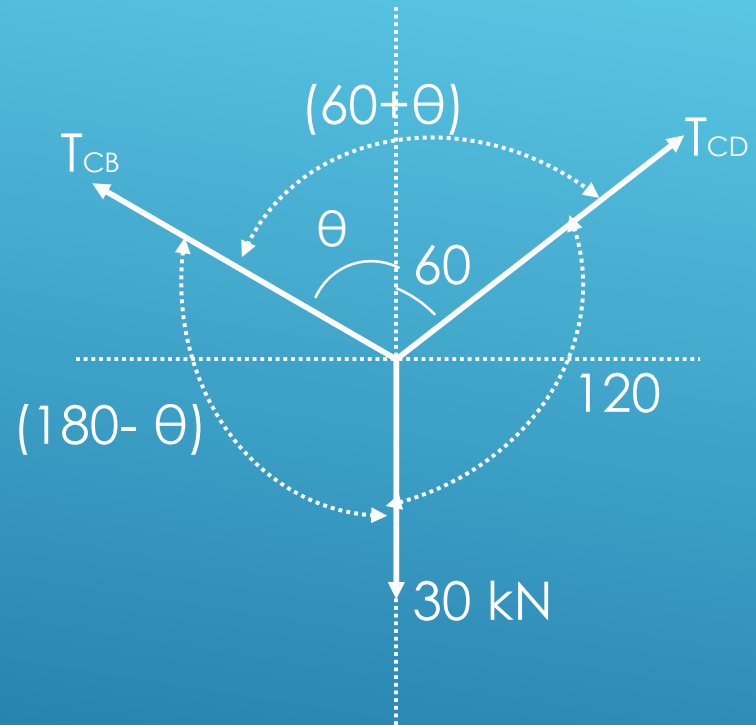
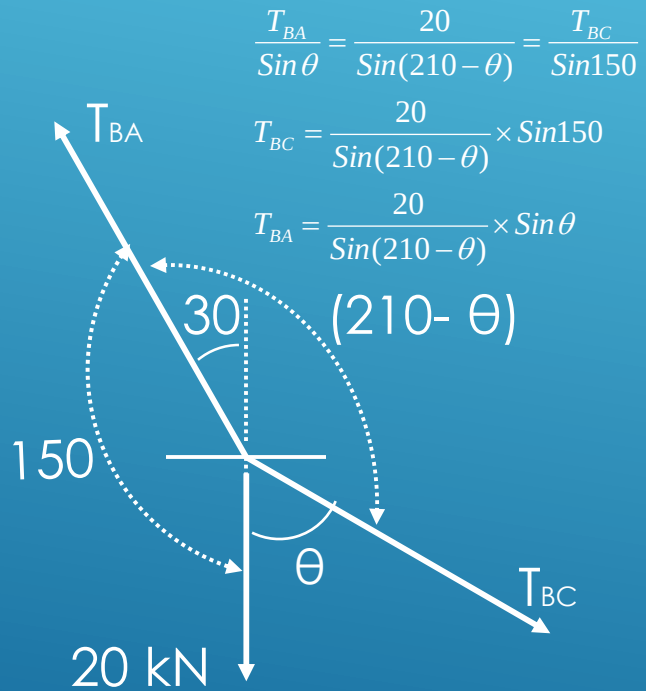


Fig (10b) FBD at Joint 'C'

Considering FBD of Joint 'B' and Using Lami's theorem,



---(1)

---(2)

Fig (10a) FBD at Joint 'B'

Considering FBD of Joint 'C' and Using Lami's theorem,

$$\frac{T_{CD}}{\sin(180 - \theta)} = \frac{25}{\sin(60 + \theta)} = \frac{T_{CB}}{\sin 120}$$

$$T_{CB} = T_{BC} = \frac{25}{\sin(60 + \theta)} \times \sin 120 \text{-----Eqn(3)}$$

$$T_{CD} = \frac{25}{\sin(60 + \theta)} \times \sin \theta \text{-----Eqn(4)}$$

P(44)

Equating R.H.S. of Eqns (1) and (3), we get,

$$\frac{20}{\sin(210 - \theta)} \times \sin 150 = \frac{25}{\sin(60 + \theta)} \times \sin 120$$

$$10(\sin 60 \cos \theta + \cos 60 \sin \theta)$$

$$= 21.65(\sin 210 \cos \theta - \cos 210 \sin \theta)$$

$$13.75 \sin \theta = 19.48 \cos \theta$$

(Continued in next slide)

(Continuation)

$$13.75 \sin \theta = 19.48 \cos \theta$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta = \frac{19.48}{13.75} \therefore \theta = 54.70^\circ$$

$$T_{BC} = T_{CB} = 23.85 \text{ kN}$$

$$T_{BA} = 38.97 \text{ kN}$$

$$T_{CD} = 22.50 \text{ kN}$$

(Continuation)

$$13.75 \sin \theta = 19.48 \cos \theta$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta = \frac{19.48}{13.75} \therefore \theta = 54.70^\circ$$

$$T_{BC} = T_{CB} = 23.85 \text{ kN}$$

$$T_{BA} = 38.97 \text{ kN}$$

$$T_{CD} = 22.50 \text{ kN}$$